A Theory of Contact and In-Group Bias*

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Abstract

Why do contact interventions durably reduce in-group bias in some settings but fail in others that appear similar? I develop a dynamic model of bias evolution through repeated social interaction. Temporary increases in out-group contact generate permanent reductions in bias only if they move society across a tipping point; otherwise, bias reverts once the intervention ends. The model identifies one mechanism through which the same intervention can generate different persistence across settings. Prevailing bias may place one population within reach of a tipping point but leave another beyond it. It also identifies a distinct role for policies that raise the value of successful interaction. By increasing the return to effort, such policies weaken the feedback that sustains high bias, temporarily eliminate the high-bias steady state, and induce convergence to the low-bias steady state. Once bias falls below the tipping point, the policy can be withdrawn without undoing the change. Temporary value-enhancing policies can therefore produce permanent change over a broader range of initial conditions than temporary contact interventions. This asymmetry implies a sequencing result. Temporary value-enhancing interventions can move a population into the region where subsequent contact interventions have lasting effects, and can shorten the duration of the value-enhancing intervention relative to relying on that intervention alone. The model also predicts a majority–minority asymmetry in long-run bias driven by differential in-group exposure.

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How can someone hate me, if they don't even know me?

- Daryl Davis

1 Introduction

Contact interventions designed to reduce in-group bias have produced a striking pattern in the empirical literature: structurally similar programs generate sharply different long-run outcomes across settings. Lowe (2021) finds that cooperative cross-caste contact through a cricket program in India generates persistent and broad reductions in discrimination that extend beyond the contact setting. By contrast, Mousa (2020) studies a closely analogous intervention in post-ISIS Iraq, pairing Christians and Muslims through a soccer program, and finds that its effects remain confined to the contact setting. More broadly, other studies find weak or null long-run effects of structured contact interventions (Scacco and Warren, 2018; Clochard et al., 2026), while meta-analyses and reviews conclude that treatment effects are highly heterogeneous across settings (Pettigrew and Tropp, 2006; Paluck et al., 2021; Lowe, 2025). A standard response is to attribute this variation to differences in implementation, population characteristics, or historical context. However, that approach neither yields a general explanation for why contact succeeds in some environments and fails in others nor provides much guidance for policy design.

This paper offers a different explanation. I develop a dynamic model of in-group bias evolution in which the long-run effect of a contact intervention depends on where the population stands relative to an endogenous tipping point. The mechanism implies that even interventions with broadly similar design can produce sharply different persistence across settings. When bias is self-reinforcing at the societal level, the dynamics admit two stable steady states, low bias and high bias, separated by an unstable threshold. A temporary increase in out-group contact generates a permanent reduction in bias only if it moves the population across that threshold; otherwise, bias declines during the intervention and reverts once contact returns to baseline. The same program can succeed in one setting and fail in another, even under similar design, because the model identifies three distinct structural channels through which settings can differ: prevailing bias may place one population within reach of the tipping point but leave another beyond it; the strength of societal feedback may generate tipping dynamics in one environment but not the other; and even when both settings exhibit tipping, differences in that feedback structure can shift the location of the tipping window so that identical initial bias falls inside it in one context and outside it in the other.

There is a population divided into two groups, a majority and a minority, whose members are repeatedly matched for social interaction.[1] The degree of assortative matching governs the frequency of out-group contact and is itself an object of policy. After matching, the probability of a positive interaction depends on three channels: the effort chosen by the matched individuals, their biases, and the aggregate level of bias in the population. Out-group interactions reduce bias when positive and reinforce it when negative, while in-group interactions reinforce bias through repeated in-group contact.[2] The state variables are the group-specific shares of biased individuals, whose co-evolution determines the long-run dynamics.

The societal feedback channel captures the idea that widespread bias shapes the quality of social interactions beyond the direct effect of the interacting individuals' own biases. When bias is pervasive, social norms and expectations can make positive out-group interaction difficult to sustain even for privately unbiased individuals, who may face external pressure to conform to the norms of their environment. When bias is rare, even privately biased individuals may find their views tempered by surrounding norms of cooperation. This feedback between aggregate bias and individual interaction outcomes is the source of the paper's tipping dynamics and has intuitive support from theories of social norm enforcement (Bowles and Gintis, 2002) and preference falsification (Kuran, 1995).

The paper's central contribution is a theory of when temporary contact generates permanent change. A temporary increase in the out-group contact rate generates a permanent reduction in bias only if the intervention is strong enough to push the population past the tipping point. If the intervention falls short, bias declines during the contact period but reverts once contact returns to baseline. This result is consistent with evidence that short-lived or weakly institutionalized contact interventions often produce weak or null effects (Paluck et al., 2019; Scacco and Warren, 2018; Enos, 2014), while sustained or structured contact can generate lasting change (Bazzi et al., 2019; Lowe, 2021; Ghosh et al., 2026). The mechanism differs from both canonical discrimination models, in which contact does not shift long-run outcomes, and belief-correction models, in which positive contact reduces bias without threshold effects (Becker, 1957; Arrow, 1973; Bordalo et al., 2016; Bohren et al., 2025).[3]

[1] Groups may correspond to latent forms of identity such as ethnicity, political orientation, race, religion, or social class. See, e.g., Akerlof and Kranton (2000, 2010), Chen and Li (2009), Bénabou and Tirole (2011), Carvalho (2016), Akerlof (2017), and Carvalho and Sacks (2021, 2024).

[2] Examples of in-group interactions reinforcing in-group bias include Tajfel et al. (1971), Tajfel and Turner (1979), Goette et al. (2006), Bernhard et al. (2006), and Chen and Li (2009).

[3] Related models include Coate and Loury (1993b), who study statistical discrimination, and Bohren et al. (2019), Pęski and Szentes (2013), and Heidhues et al. (2026), who study belief distortions and social norms. In these settings, contact operates through informational or reputational channels rather than through tipping dynamics, and does not deliver the sequencing result below.

Where contact interventions fail, increasing the value of successful interaction can succeed. In the model, this is captured by an increase in V, the payoff from a successful interaction. Thus, V should be interpreted not as contact per se, but as the material or institutional stake attached to successful interaction. Policies that raise V therefore correspond to interventions that make successful interaction more valuable, including economic integration into high-value productive settings, team-based pay, or school environments organized around meaningful shared goals. By raising the return to effort, such policies induce investment in positive interactions even in high-bias environments and thereby weaken the feedback loop that sustains the high-bias steady state. Because long-run bias is shaped by the success or failure of out-group interactions, a higher V reduces bias by making those interactions more likely to succeed. When the returns to successful interaction are sufficiently high, the high-bias steady state disappears during the intervention, leaving a unique low-bias steady state to which the population converges from any initial condition. Once bias falls below the original tipping point, the policy can be withdrawn without undoing the change. Unlike contact interventions, value-based interventions can therefore generate permanent change even in settings where temporary increases in contact would fail.

The asymmetry between the two instruments implies a sequencing result. The instruments are complements rather than substitutes. A sufficiently strong value-based intervention alone can generate permanent change from any initial condition, but may require a prolonged or intensive intervention. A natural sequencing logic therefore combines the two instruments. A temporary value-based intervention first reduces bias enough to move the population into the tipping region, after which a temporary increase in out-group contact can push it across the threshold that contact interventions alone could not have reached. The formal sequencing result further shows that such a policy sequence can shorten the duration of the value-based intervention relative to a value-only policy, though not necessarily the total duration of intervention. Once the economy crosses the baseline tipping point, both interventions can be withdrawn and the low-bias outcome becomes self-sustaining.

This interpretation is consistent with evidence from high-value integration settings. Rao (2019) documents persistent reductions in discriminatory behavior among wealthy students following integration with low-income peers in elite Delhi schools, with the largest effects appearing in economic exchange tasks where interaction stakes are highest. Boisjoly et al. (2006) find lasting attitude change among white students randomly assigned African American roommates, and Beaman et al. (2009) find persistent reductions in gender bias following repeated exposure to female leaders under a quota system. In each case, the intervention contains features that can be interpreted through the model as raising the value of successful interaction in settings organized around out-group cooperation, rather than merely increasing contact rates.[4]

The effort margin creates a central tension in the model. Effort reduces steady-state bias while weakening the feedback required for permanent regime change. Because individuals respond to rising aggregate in-group bias by investing more effort in out-group interactions, the feedback loop is dampened relative to a model without effort. This attenuation has two consequences. First, effort rules out the fully biased steady state. Even in high-bias environments, individuals invest enough in out-group interactions to keep universal bias from being sustained. Second, effort shrinks the parameter region supporting multiple steady states, so the conditions required for tipping dynamics are harder to satisfy than a model without effort would suggest. Raising the value of successful interaction therefore lowers steady-state bias by inducing investment in effort. However, the same response flattens the feedback and narrows the conditions under which tipping dynamics arise. This tension is absent from standard models of evolutionary dynamics and equilibrium selection (Schelling, 1971; Kandori et al., 1993; Young, 1993). The same force that reduces steady-state bias also attenuates the feedback required for temporary interventions to generate permanent change. The model therefore yields a sharp empirical prediction. Environments with higher stakes for successful interaction should exhibit lower steady-state bias, but a smaller region in which temporary increases in contact generate lasting change. Card et al. (2008) show that tipping points play an important role in the dynamics of residential segregation; the present paper identifies an analogous threshold structure in the evolution of in-group bias, but shows that effort makes that structure harder to sustain. This attenuation result is central to the paper. Adaptation to bias via effort does not merely reduce bias in levels, it also changes which policies can generate permanent change.

The model also yields a sharp asymmetry in steady state in-group bias. Under empirically plausible conditions of non-fully-assortative matching and positive in-group reinforcement, majority members are strictly more biased than minority members in any interior steady state, and minority members invest more effort in out-group interactions than majority members. The mechanism is differential in-group exposure. Because the majority's in-group interaction rate exceeds the minority's, majority members accumulate more in-group reinforcement, which shifts their steady-state bias upward. A uniform increase in out-group contact reduces this gap, generating convergence in the distribution of

[4] This is consistent with Allport's (1958) argument that contact is most effective when it occurs under cooperative goals and meaningful shared stakes.

bias across groups alongside a reduction in average bias, consistent with evidence that contact effects are often stronger among majority than minority group members (Tropp and Pettigrew, 2005; Ghosh et al., 2026), and with evidence that higher contact intensity can backfire among majority but not minority participants (Ghosh et al., 2026). This asymmetry is distinct from that in identity-based (Akerlof and Kranton, 2000; Shayo, 2009) or cultural-transmission and homophily models (Bisin and Verdier, 2001; Currarini et al., 2009; Golub and Jackson, 2012), because here it emerges from the interaction structure itself rather than from differences in preferences or socialization across groups.

Relative to existing models of social interactions with aggregate feedback, the paper's contribution is not simply to show that sufficiently nonlinear social feedback can generate tipping. Rather, it identifies how tipping dynamics are reshaped once out-group interaction is mediated by endogenous effort and match-specific bias. The model delivers three results that do not arise from the societal-feedback channel alone. First, endogenous effort attenuates the aggregate feedback that sustains multiplicity, so the same force that lowers steady-state bias also makes permanent regime change harder to achieve. Second, temporary contact and value-enhancing interventions operate through distinct mechanisms. The former works, when it works, by shifting the tipping point and changing basins of attraction, whereas the latter can eliminate the high-bias steady state altogether by raising the return to effort. Third, this asymmetry implies a formal sequencing result. A temporary value-enhancing intervention can move the economy into the region where contact becomes self-sustaining, and can shorten the required duration of the value-enhancing phase relative to relying on that intervention alone. In the effort-only benchmark, the dynamics admit a unique steady state. In the societal-feedback benchmark, sufficiently S-shaped feedback generates multiplicity and tipping. In the full model, endogenous effort dampens the aggregate feedback that sustains multiplicity, shrinking the region in which temporary increases in contact can generate permanent change. As V rises, this attenuation becomes stronger, and the multiplicity region can disappear altogether. Figure 5 illustrates these results.

The remainder of the paper is structured as follows. Section 2 develops the model and establishes preliminary results. Sections 3 and 4 form the paper's analytical core. The former isolates each mechanism individually, while the latter combines them to characterize the full dynamics. Section 5 derives policy implications, and Section 6 concludes.

2 The Model

Time is discrete and indexed by $t = 0, 1, 2, \ldots$. There is a fixed population of individuals $I = [0, 1]$ with Lebesgue measure λ. Each $i \in I$ is born with a type $\tau \in \{1, 2\}$. $I_\tau \subset I$ denotes the subset of τ types. The share of type 1 individuals is $q = \lambda(I_1) > \frac{1}{2}$, so type 1 individuals constitute the majority. Generically, $q_\tau = \lambda(I_\tau)$. Each $i \in I$ is also endowed with an in-group bias parameter $b_i^t \in \{0, 1\}$. If $b_i^t = 0$, then i is *unbiased* and if $b_i^t = 1$, then i is *biased.* These biases evolve over time in a manner described below. Let $\pi_\tau^t = \frac{1}{\lambda(I_\tau)} \int_{I_\tau} b_i^t d\lambda \in [0, 1]$ denote the share of biased type τ individuals in the population at time t and $\pi^t = (\pi_1^t, \pi_2^t)$. The initial state π^0 is exogenously given. Let $\bar{\pi}^t = q\pi_1^t + (1 - q)\pi_2^t$ denote the aggregate prevalence of in-group bias in the population. When there is no confusion, I suppress the time notation.

Timing. Social interactions in each period take place as follows.

(1) Each individual i is matched with an individual j, at which point (τ_i, b_i) is observed by j and (τ_j, b_j) is observed by i. Let

$$\eta_\tau = \lambda(I_\tau) - \mu(1 - \lambda(I_\tau))$$

denote the probability that a type τ individual is matched with another type τ individual.[5] The *match parameter* $\mu \in \left[-1, \frac{1-q}{q}\right]$ guarantees that $\eta_\tau \in [0, 1]$ for $\tau = 1, 2$. Random matching occurs when $\mu = 0$, assortative matching occurs when $\mu < 0$, and disassortative matching occurs when $\mu > 0$.

(2) Upon individuals i and j being matched, each chooses an effort level $e \in [0, 1]$ at cost $\frac{1}{2}e^2$ to maximize current utility, described below.[6] Their efforts e_i and e_j are aggregated according to the function

$$s(e_i, e_j) = e_i + e_j - e_i e_j, \tag{1}$$

which determines the probability that effort alone yields a positive interaction, described next.[7]

[5] This specification satisfies market clearing: $q(1 - \eta_1) = (1 - q)(1 - \eta_2)$; i.e., the share of type 1 individuals matched with type 2 individuals must equal the share of type 2 individuals matched with type 1 individuals.

[6] The qualitative results are invariant to whether effort is chosen before or after matching. Under pre-match timing, individuals form expectations over the type-bias distribution of their match, yielding a Bayesian Nash equilibrium characterized by a 4×4 linear system with the same properties as Proposition 1.

[7] The specific functional form $s(e_i, e_j) = e_i + e_j - e_i e_j$ is adopted for tractability rather than necessity. The results extend to any function $s : [0, 1]^2 \to [0, 1]$ that is increasing and concave in each player's own effort with effort choices as strategic substitutes. Under these conditions, equilibrium effort is increasing in $(1 - \beta(b_i, b_j))\theta(\bar{\pi})V$,

(3) The social interaction between i and j occurs. Three factors determine whether this interaction is positive or negative.

   (i) *Effort.* With probability $s(e_i, e_j)$, the interaction is positive.

   If effort is not sufficient to yield a positive interaction, then the biases of the matched individuals and the broader social environment influence the outcome.

   (ii) *Individual feedback.*

   If effort fails to yield a positive interaction, occurring with probability $1 - s(e_i, e_j)$, then the matched individuals' biases determine the probability that the interaction is resolved positively at the next stage. Given biases b_i and b_j, let

$$\beta(b_i, b_j) = \begin{cases} \beta_0 & \text{if } b_i + b_j = 0 \\ \beta_1 & \text{if } b_i + b_j = 1 \\ \beta_2 = 0 & \text{if } b_i + b_j = 2, \end{cases} \tag{2}$$

   where $0 = \beta_2 < \beta_1 < \beta_0 \leq 1$.[8]

   For in-group matches, I suppress the bias notation and treat the matched pair as if $(b_i, b_j) = (0, 0)$, since the individual-bias channel is only relevant for out-group interactions.

   (iii) *Societal feedback.* If neither effort nor overcoming individual biases are sufficient to yield a positive interaction, occurring with probability $(1 - s(e_i, e_j))(1 - \beta(b_i, b_j))$, the outcome of the interaction is determined by the broader social environment. Let $\bar{\pi}$ denote the average level of bias in the population. Conditional on reaching this stage, the probability that the interaction fails is given by $\theta(\bar{\pi})$, where $\theta : [0, 1] \to [0, 1]$ is

which is the only way s enters the equilibrium comparative statics and the fixed-point structure of the dynamics. The assumption that effort choices are strategic substitutes reflects the standard free-riding incentive in team and pair-based production (Holmström, 1982). This interpretation is consistent with field evidence showing that team productivity in socially divided workplaces depends sensitively on how incentives are structured (Hjort, 2014).

[8] $\beta_0 > 0$ implies that, even in the absence of effort, unbiased individuals extend some baseline trust and willingness to engage in interaction (Berg et al., 1995; Glaeser et al., 2000). The assumption $\beta_1 < \beta_0$ reflects the idea that bias reduces the probability of successful coordination by distorting expectations and incentives in out-group interaction (Arrow, 1973; Akerlof and Kranton, 2000). Finally, $\beta_2 = 0$ captures a limiting case in which mutual bias produces complete distrust and interaction breakdown, consistent with theories of out-group prejudice and discrimination (Tajfel and Turner, 1979; Becker, 1957; Coate and Loury, 1993a). Related models show how beliefs about group characteristics can generate persistent discrimination and inefficient interaction outcomes (Arrow, 1973; Coate and Loury, 1993b; Bohren et al., 2019, 2025). $\beta_2 = 0$ is adopted for convenience, not necessity. For any $\kappa \in (0, \beta_1)$ $\rho_{1,1} > 0$ and the fully biased steady state shifts by continuity to an interior point at distance $O(\kappa)$ from $(1, 1)$. All qualitative results are unchanged.

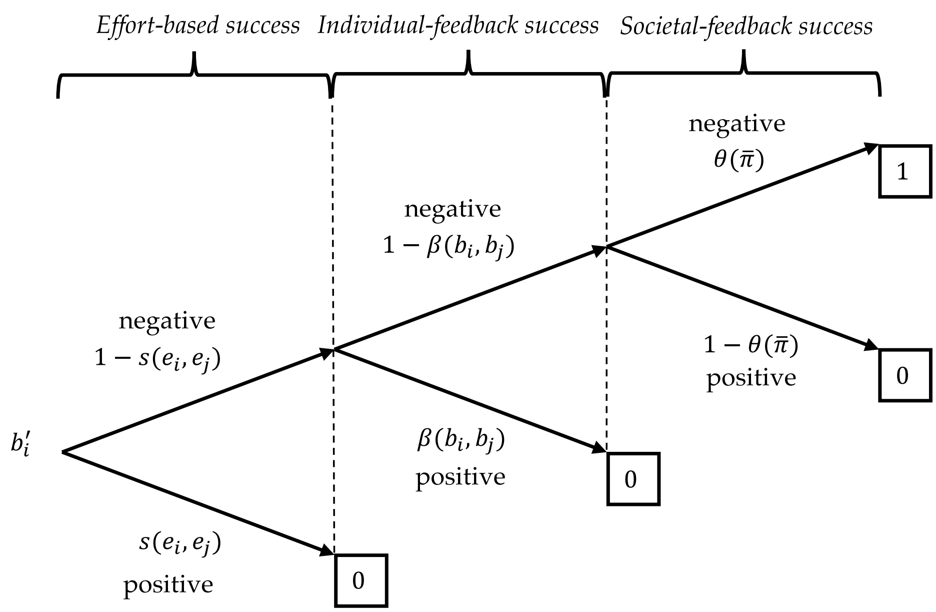


Figure 1: Diagrammatic representation of the probability of a positive interaction and future bias $b_{i'}$ given current bias b_i and the matched individual's bias b_j. *Positive* corresponds to a positive interaction and *negative* to a negative interaction.

increasing and twice continuously differentiable with $\theta(0) = 0$ and $\theta(1) = 1$. I interpret $\theta(\bar{\pi})$ as a reduced form model of societal feedback that captures the idea that when bias is widespread, social norms and expectations make successful interaction less likely even after conditioning on the characteristics of the interacting pair. Section 2.1 offers two microfoundations that yield this reduced form specification.

Note that the paper's results are invariant to the ordering of (i)–(iii).[9]

(4) Biases are updated as follows. If $\tau_i = \tau_j$ at time t, then with probability $\varepsilon > 0$, $b_i^{t+1} = 1$ and with complementary probability, $b_i^{t+1} = b_i^t$. If $\tau_i \neq \tau_j$ at time t, then a positive interaction yields $b_i^{t+1} = 0$ and a negative interaction yields $b_i^{t+1} = 1$.[10]

Hence, conditional on i and j being matched, the probability of a positive interaction is

$$\rho_{b_i,b_j}(e_i, e_j) = s(e_i, e_j) + (1 - s(e_i, e_j))[\beta(b_i, b_j) + (1 - \beta(b_i, b_j))(1 - \theta(\bar{\pi}))]. \qquad (3)$$

This process is illustrated in Figure 1.

[9] As $\rho_{b_i,b_j}(e_i, e_j) = s(e_i, e_j) + (1 - s(e_i, e_j))[\beta(b_i, b_j) + (1 - \beta(b_i, b_j))(1 - \theta(\bar{\pi}))]$, which can be written as $1 - (1 - s)(1 - \beta)\theta$, all six orderings of the three channels are identical (as this probability is symmetric in effort, individual feedback, and societal feedback).

[10] I discuss the construction of ε in greater detail in Section 2.2.

Payoffs. Conditional on being matched with an individual j with type-bias pair (τ_j, b_j), individual i's utility is

$$u(e_i, e_j; (\tau_i, b_i), (\tau_j, b_j)) = \rho_{b_i,b_j}(e_i, e_j)V - \frac{1}{2}e_i^2, \tag{4}$$

where $V > 0$ is the value received from a successful interaction. Throughout, a positive interaction is the payoff-relevant successful outcome of the interaction. A high V implies high interaction stakes and a low V implies low interaction stakes.

Equilibrium Definitions. At each time t, the utility-maximizing type-and-bias contingent effort levels constitute a Nash equilibrium (NE) of the social interaction stage game if, for all i, $e(\tau_i, b_i) = \arg\max_{e_i \in [0,1]} u(e_i, e(\tau_j, b_j); \tau_i, b_i, \tau_j, b_j)$. This equilibrium describes within-period strategic interactions. Across periods, the distribution of biases evolves according to the process described above.

Let $\pi^\infty = (\pi_1^\infty, \pi_2^\infty)$ denote the steady-state shares of in-group bias among each type and $\bar{\pi}^\infty$ the steady-state aggregate bias. I define two steady states of interest. The biased steady state is the steady state in which the entire population is biased ($\bar{\pi}^\infty = 1$), and the unbiased steady state is the steady state in which the entire population is unbiased ($\bar{\pi}^\infty = 0$).

2.1 Microfoundations of the Societal-Feedback Channel

While the model treats $\theta(\bar{\pi})$ as a reduced-form object, it can backed out from structural models of societal interactions. The first discussed here is community norm enforcement. Conditional on reaching the societal-feedback stage, the probability of a successful interaction is mediated by the biases of a finite reference group drawn from the population.

Consider a model of community norm enforcement. Conditional on reaching the societal-feedback stage, the interaction is mediated by a reference group of k members drawn independently from the population. If a majority of the k members drawn are biased, then the social interaction is deemed a failure and if a majority are unbiased, then the social interaction is successful. Thus,

$$\theta(\bar{\pi}) = \Pr\left(\text{Binomial}(k, \bar{\pi}) > \frac{k}{2}\right) = \sum_{j=\lfloor k/2 \rfloor + 1}^{k} \frac{k!}{j!(k-j)!}\bar{\pi}^j(1-\bar{\pi})^{k-j}.$$

For all k, $\theta(0) = 0$ and $\theta(1) = 1$. At $k = 1$, $\theta(\bar{\pi}) = \bar{\pi}$ and at $k = 2$, $\theta(\bar{\pi}) = \bar{\pi}^2$.[11]

[11] For even k, ties arise with positive probability. Any tie-breaking rule preserves the qualitative conclusions emphasized here. $\theta(\bar{\pi})$ remains increasing, $k = 1$ yields linear feedback, $k = 2$ yields convex feedback, and larger k produces increasingly sharp threshold behavior. The exact location of the inflection point may depend on how ties are resolved, but none of the analytical results relies on that detail.

For all $k \geq 3$, θ S-shaped, and $\sup_{\bar{\pi}} \frac{d\theta(\bar{\pi})}{d\bar{\pi}}$ is strictly increasing in k. Thus, larger reference groups generate sharper threshold behavior in societal feedback. Moreover, as $k \to \infty$, $\theta(\bar{\pi})$ converges pointwise to a step function at $\bar{\pi} = \frac{1}{2}$, so the size of the relevant social reference group governs the strength of societal feedback. The formal link between this curvature and the existence of multiple steady states is developed below.[12]

A preference-falsification environment in the spirit of Kuran (1995) can also generate the reduced form specification above.These microfoundations capture the broader idea, emphasized in the literature on trust and social capital, that the social environment shapes the expressed quality of bilateral interactions (Bowles and Gintis, 2002). These microfoundations are illustrative rather than essential. They offer structural environments in which the reduced-form societal-feedback channel can arise, but none of the analytical results depends on any specific functional form or underlying mechanism of social interaction. The proofs are stated for a general reduced-form $\theta(\bar{\pi})$.

2.2 Preliminary Results

Before characterizing the full dynamics, I establish three preliminary results. I first solve for the effort levels that constitute a Nash equilibrium of the social interaction stage game. I then characterize the implied average effort gap between minority and majority individuals. Lastly, I highlight the role of randomness and in-group reinforcement in the evolution of bias.

Recall that, conditional on being matched with a type τ_j individual with bias b_j, individual i chooses the effort level that maximizes (4).

Proposition 1. *Suppose that pairs (τ_i, b_i) and (τ_j, b_j) are matched. The effort levels constituting an Nash equilibrium of the social interaction stage game are given by*

$$e_i^*(b_i, b_j) = \frac{(1 - \beta(b_i, b_j))\theta(\bar{\pi})V}{1 + (1 - \beta(b_i, b_j))\theta(\bar{\pi})V}. \tag{5}$$

Thus, effort levels are increasing in the value of a successful interaction, the match-specific probability of failure, and the societal-feedback function $\theta(\bar{\pi})$.

The proofs of this and all other results are in the Appendix. Proposition 1 implies that equilibrium effort is increasing in the match-specific probability of a negative interaction, $1 - \beta(b_i, b_j)$, the

[12] The conditions for multiple steady-states are outlined in Proposition 7. When the remaining conditions of Proposition 7 are satisfied, there always exists a k large enough to generate multiple steady states and tipping dynamics.

societal-feedback term $\theta(\bar{\pi})$, and the value of a successful interaction V. Interpreting effort as costly preventive investment, individuals invest more when baseline success is less likely. This logic is consistent with evidence that cooperation rises when punishment or exclusion makes opportunism more consequential (Fehr and Gächter, 2000; Cinyabuguma et al., 2005; Gürerk et al., 2006). Related field evidence shows that incentive design, including team-based pay, can mitigate productivity losses associated with inter-group divisions (Hjort, 2014).

Proposition 1 also allows a comparison of average effort between minority (type 2) and majority (type 1) individuals. Given (5), the average effort put forth by a type τ is

$$(1-\eta_\tau)[\pi_1\pi_2 e^*(1,1) + ((1-\pi_1)\pi_2 + (1-\pi_2)\pi_1)e^*(1,0) + (1-\pi_1)(1-\pi_2)e^*(0,0)] + \eta_\tau e^*(0,0),$$

where $e_i^*(b_i, b_j) = e^*(b_i, b_j)$ as i enters only through b_i and $e^*(1,0) = e^*(0,1)$.

Corollary 1. *Average effort is weakly higher among minority individuals than majority individuals, strictly so when $\mu > -1$ and in-group bias is present in the population ($\bar{\pi} > 0$).*

2.2.1 The Role of Randomness and In-Group Reinforcement

Let $s^*(b_i, b_j) \equiv s(e_i^*(b_i, b_j), e_j^*(b_i, b_j))$ denote the aggregate effort in the NE of the social interaction stage game. Now, (3) can be written as a function only of biases b_i, b_j:

$$\rho_{b_i,b_j} = s^*(b_i, b_j) + (1 - s^*(b_i, b_j))[\beta(b_i, b_j) + (1 - \beta(b_i, b_j))(1 - \theta(\bar{\pi}))].$$

For a randomly drawn type τ individual, the expected probability of a positive interaction is

$$\bar{\rho}_\tau(\pi) = (1-\pi_\tau)\left[(1-\pi_{-\tau})\rho_{0,0} + \pi_{-\tau}\rho_{0,1}\right] + \pi_\tau\left[(1-\pi_{-\tau})\rho_{1,0} + \pi_{-\tau}\rho_{1,1}\right],$$

which implies that

$$\pi_\tau^{t+1} = (1-\eta_\tau)\left(1 - \bar{\rho}_\tau(\pi^t)\right) + \eta_\tau\left(\pi_\tau^t + (1-\pi_\tau^t)\varepsilon\right)$$

Hence,

$$\Delta\pi_\tau = (1-\eta_\tau)\left[1 - \bar{\rho}_\tau(\pi) - \pi_\tau\right] + \eta_\tau(1-\pi_\tau)\varepsilon. \tag{6}$$

The role of randomness ($\varepsilon > 0$) is immediately apparent. It is necessary for out-group contact to affect steady-state bias in the population. When $\varepsilon = 0$, out-group interaction, and thus changes in the frequency of out-group contact, affect only the speed of adjustment, not the long-run level of bias.

I interpret $\eta_\tau(1-\pi_\tau)\varepsilon$ as a reduced-form socialization or identity-reinforcement mechanism. Conditional on in-group interaction, a fraction ε of initially unbiased individuals adopt biased beliefs or norms through exposure to in-group narratives, conformity incentives, and local transmission. This interpretation is consistent with economic models of cultural transmission in which in-group socialization and assortative matching yield persistent group-specific attitudes in the population (Bisin and Verdier, 2000, 2001; Carvalho and Sacks, 2024; Carvalho et al., 2024a).[13] Identity-based frameworks similarly imply that social categories and moral identity shape behavior and beliefs, and that in-group interactions can sustain and reinforce group-consistent worldviews (Akerlof and Kranton, 2000; Bénabou and Tirole, 2011). Because homophily and segregation raise the frequency of in-group interactions, even small in-group reinforcement rates can have quantitatively important implications for long-run bias (Currarini et al., 2009; Gentzkow and Shapiro, 2011).

3 Mechanism Decomposition

Before analyzing the full model, I decompose it into its three constituent mechanisms: effort, individual feedback, and societal feedback, and study the long-run behavior that each generates in isolation. For each channel, I characterize the steady states and convergence as if that channel were the only force. This decomposition clarifies which aspects of the dynamics operate through each channel.

3.1 Effort

I begin with the effort channel in isolation. Can effort alone can generate persistent effects under increased out-group contact? It cannot. Although effort responds to and shapes the probability of a positive interaction, effort admits a unique globally stable steady state. Therefore, temporary changes in μ or V are only transitory. Once μ or V reverts, so too does the steady state.

Fix $\beta(b_i, b_j) = 0$ and $\theta(\bar{\pi}) = 1$ for all b_i, b_j, and $\bar{\pi}$. In this case, the probability of a positive interaction is $\rho_{b_i,b_j}(e_i, e_j) = s(e_i, e_j)$. By (5),

$$e_i^* = \frac{V}{1+V} \equiv e^*.$$

Evaluating the probability of a positive interaction at (e^*, e^*) yields

$$s(e^*, e^*) = \frac{V(2+V)}{(1+V)^2} \equiv s^* \in (0,1).$$

[13] See Shayo (2020) for a survey of empirical and experimental evidence on in-group bias and in-group conformity.

Thus, when $V > \sqrt{2} - 1$, a positive interaction more likely than a negative one and vice versa.

The evolution of each π_τ is given by

$$\Delta\pi_\tau = (1 - \eta_\tau)\left[(1 - \pi_\tau)(1 - s^*) - \pi_\tau s^*\right] + \eta_\tau(1 - \pi_\tau)\varepsilon, \tag{7}$$

which yields the unique steady-state values

$$\pi_\tau^\infty = \begin{cases} 1 - \frac{(1-\eta_\tau)s^*}{1-\eta_\tau(1-\varepsilon)} & \text{if } \varepsilon > 0 \text{ or } \mu > -1 \\ \pi_\tau^0 & \text{if } \varepsilon = 0 \text{ and } \mu = -1. \end{cases} \tag{8}$$

If $\varepsilon = 0$ and $\eta_\tau = 1$ (i.e., $\mu = -1$), matching is purely assortative and biases remain constant.[14] The following proposition characterizes convergence and the comparative statics of the steady state with respect to the match parameter μ and the interaction value V.

Proposition 2. *Fix $\beta(b_i, b_j) = 0$ and $\theta(\bar{\pi}) = 1$ for all b_i, b_j, and $\bar{\pi}$. The following statements hold.*

(i) *The dynamics converge to* (8) *from all initial $\pi^0 \in [0,1]^2$.*

(ii) *If either $\varepsilon = 0$ and $\mu > -1$ or $\varepsilon > 0$ and $\mu = -1$, then $\pi_1^\infty = \pi_2^\infty$. If $\varepsilon = 0$ and $\mu = -1$, then $\pi_\tau^\infty = \pi_\tau^0$. Otherwise, $\pi_1^\infty > \pi_2^\infty$.*

(iii) *For all $\varepsilon \in (0,1]$, π_τ^∞ is strictly decreasing in μ. For all $\varepsilon \in [0,1]$ and $\mu > -1$, π_τ^∞ is strictly decreasing in V.*

Proposition 2 delivers a clean comparative static. When $\varepsilon > 0$, increasing out-group contact (higher μ) strictly reduces steady-state bias. Increased contact reduces the weight on in-group interactions and increases the frequency of out-group encounters in which costly effort can generate positive interaction. The effect is amplified by V, as higher-value interactions increase effort and the corresponding probability of success.

This mechanism aligns with the canonical contact hypothesis, which emphasizes the role of structured, high-stakes cooperation in improving out-group attitudes (Allport, 1958; Pettigrew and Tropp, 2006), and with field evidence showing that sustained collaborative exposure reduces prejudice and discrimination (e.g., Lowe, 2021; Scacco and Warren, 2018; Bazzi et al., 2019). A temporary increase in contact shifts the transition path but not the steady state distribution. Once μ returns to its original value, so too does the steady state. The same logic holds for V.

[14] These statements are verified in the proof of Proposition 2.

This is consistent with evidence that the effects of short-lived contact interventions attenuate once contact ends. Absent continued interaction, the effects frequently decay over time (Paluck et al., 2021; Macchiavello and Morjaria, 2015). Permanent effects, by contrast, tend to arise in settings where out-group interaction is sustained or integrated into institutions (Bazzi et al., 2019).

3.2 Individual Feedback

I next isolate the individual-bias channel by shutting down effort and societal feedback. Unlike the effort-only benchmark, the model can now admit biased, unbiased, or interior steady states depending on parameters. However, for any fixed parameter configuration and $\mu > -1$, the dynamics converge to a unique steady state. Like the effort benchmark, temporary changes in out-group contact therefore affect only the transition path, not the steady state biases. V has no effect, as it only operates through effort.

Fix $s(e_i, e_j) = 0$ and $\theta(\bar{\pi}) = 1$ for all e_i, e_j, and $\bar{\pi}$. In this case, the probability of a positive interaction is given by $\rho_{b_i,b_j}(e_i, e_j) = \beta(b_i, b_j)$ and the evolution of π_τ is given by

$$\Delta\pi_\tau = (1-\eta_\tau)\Big[(1-\pi_\tau)\big((1-\pi_{-\tau})(1-\beta_0) + \pi_{-\tau}(1-\beta_1)\big) - \pi_\tau(1-\pi_{-\tau})\beta_1\Big] + \eta_\tau(1-\pi_\tau)\varepsilon. \quad (9)$$

Setting $\Delta\pi_\tau = 0$ and solving for π_τ yields the steady-state mapping

$$F_\tau(\pi_{-\tau};\varepsilon) = \frac{(1-\eta_\tau)[(1-\pi_{-\tau})(1-\beta_0) + \pi_{-\tau}(1-\beta_1)] + \eta_\tau\varepsilon}{(1-\eta_\tau)[(1-\pi_{-\tau})(1-\beta_0+\beta_1) + \pi_{-\tau}(1-\beta_1)] + \eta_\tau\varepsilon}, \quad (10)$$

so

$$\frac{\partial F_\tau(\pi_{-\tau};\varepsilon)}{\partial\pi_{-\tau}} = \frac{(1-\eta_\tau)\beta_1[(1-\eta_\tau)(1-\beta_1) + \eta_\tau\varepsilon]}{((1-\eta_\tau)[(1-\pi_{-\tau})(1-\beta_0+\beta_1) + \pi_{-\tau}(1-\beta_1)] + \eta_\tau\varepsilon)^2}. \quad (11)$$

The following proposition characterizes the steady states implied by (9) and (11).

Proposition 3. *Fix $s(e_i, e_j) = 0$ and $\theta(\bar{\pi}) = 1$ for all e_i, e_j, and $\bar{\pi}$. The biased steady state always exists. If $\varepsilon = 0$ and $\mu > -1$, then the following statements hold.*

(i) *Almost all initial states converge to the biased steady state iff either $\beta_1 < \frac{1}{2}$ or $\beta_1 = \frac{1}{2}$ and $\beta_0 < 1$.*

(ii) *If $\beta_0 = 1$, then an unbiased steady state exists, to which almost all initial states converge iff $\beta_1 > \frac{1}{2}$.*

(iii) *If* $\beta_1 > \frac{1}{2}$ *and* $\beta_0 < 1$*, then there exists a unique interior steady state to which almost all states converge. If* $\beta_1 = \frac{1}{2}$ *and* $\beta_0 = 1$*, then each initial state converges to* $\pi_\tau^t = q\pi_1^0 + (1-q)\pi_2^0$.

If $\varepsilon > 0$ *and* $\mu > -1$*, then the following statements hold.*

(iv) *If* $\frac{\partial F_1(1;\varepsilon)}{\partial \pi_2} \times \frac{\partial F_2(1;\varepsilon)}{\partial \pi_1} \leq 1$*, then the biased steady state is the unique steady state.*

(v) *If* $\frac{\partial F_1(1;\varepsilon)}{\partial \pi_2} \times \frac{\partial F_2(1;\varepsilon)}{\partial \pi_1} > 1$*, then there exists an interior (asymmetric) steady state, which is componentwise strictly increasing in* ε*, to which almost all initial conditions converge.*

If $\mu = -1$*, then* $\pi_\tau^\infty = 1$ *when* $\varepsilon > 0$ *while* $\pi_\tau^\infty = \pi_\tau^0$ *when* $\varepsilon = 0$.

When $\varepsilon = 0$ and $\mu > -1$, the steady states are determined by the relative values of β_0 and β_1. If $\beta_1 \leq \frac{1}{2}$, out-group interactions are not sufficiently bias-reducing and the biased steady state attracts almost all initial conditions. If $\beta_1 > \frac{1}{2}$, the dynamics converge to an interior steady state whenever $\beta_0 < 1$, given by

$$\pi_1^\infty = \pi_2^\infty = \frac{1-\beta_0}{2\beta_1 - \beta_0}, \tag{12}$$

and to the unbiased steady state when $\beta_0 = 1$. Introducing $\varepsilon > 0$ eliminates the unbiased steady state and shifts any interior steady state upward. Full bias becomes globally stable whenever out-group updating near $(\pi_1, \pi_2) = (1,1)$ is locally too weak to offset in-group reinforcement, as stated in part (iv).

Two remarks about the benchmark case $\varepsilon = 0$ are in order. First, except under full assortativity ($\mu = -1$), steady states are independent of the match parameter μ. Varying μ affects the speed of convergence but not the steady state. Second, a biased steady state always exists.[15] When $\beta_1 \leq \frac{1}{2}$, out-group interactions either reduce bias only weakly or reinforce it, and the population converges to full bias. In this region, increasing out-group contact increases the speed at which bias accumulates rather than mitigating it. This feature is consistent with empirical evidence showing that contact reduces prejudice primarily when interactions are cooperative and generate positive informational signals, and may instead entrench bias when interactions are conflict-oriented or unstructured (Allport, 1958; Pettigrew and Tropp, 2006; Lowe, 2021; Enos, 2014).

When $\varepsilon > 0$, the dynamics incorporate in-group reinforcement in addition to out-group updating. Even when out-group interactions are mildly positive, in-group contact generates drift toward bias.

[15] This follows from $\rho_{1,1}(0,0) = 0$. When two biased individuals are matched, the probability of a positive interaction (absent effort) is zero, so there is no endogenous force reducing bias at $(1,1)$.

The unbiased steady state therefore disappears except under full assortativity, and any interior steady state shifts upward. Long-run outcomes are determined by the balance between positive bias-reducing out-group interactions and bias reinforcement near the fully biased state. The slope condition in part (iv) formalizes this trade-off. Full bias becomes globally stable whenever out-group updating is locally too weak to offset in-group reinforcement.

This mechanism resonates with models of cultural transmission and identity formation in which in-group socialization sustains persistent group-specific beliefs (Bisin and Verdier, 2000, 2001; Akerlof and Kranton, 2000; Bénabou and Tirole, 2011; Carvalho and Sacks, 2024). Recent empirical evidence further suggests that contact alone is insufficient to reduce polarization when informational environments amplify in-group narratives. Studies of media consumption and online platforms document how selective exposure and social reinforcement can entrench partisan attitudes even amid out-group interaction (Gentzkow and Shapiro, 2011; Allcott and Gentzkow, 2017; Boxell et al., 2017). Experimental evidence likewise shows that out-group interactions may have limited or even backfiring effects when interactions trigger defensive identity responses (Bail et al., 2018).[16]

Because the in-group interaction rates η_1 and η_2 generally differ (except under full assortativity, $\mu = -1$), the interior steady state typically features asymmetric bias across types when $\varepsilon > 0$. This raises a natural comparative question: when there is an interior steady state, is the majority more or less biased than the minority in the long run?

Corollary 2. *Fix $\mu > -1$ and suppose $\varepsilon > 0$. Under the conditions of Proposition 3 such that an interior steady state exists, $\pi_1^\infty > \pi_2^\infty$.*

For any $\mu > -1$, the majority's in-group interaction rate exceeds that of the minority's: $\eta_1 - \eta_2 = (2q-1)(1+\mu) > 0$, so the majority experiences less out-group contact than the minority. When $\varepsilon > 0$, within-type interactions generate drift toward bias, so a higher η_τ shifts type τ's steady-state response upward. As a consequence, whenever the limiting state is interior, the majority ends up strictly more biased than the minority.

Relatedly, when $\varepsilon > 0$, increasing μ generally lowers steady-state bias. Because in-group reinforcement operates through η_τ, a rise in μ weakens the drift toward bias and strengthens the relative force of out-group contact-based bias reduction. As a result, parameters for which the biased steady

[16] Additional theoretical evidence includes Carvalho, Koyama and Sacks (2017) and Carvalho et al. (2024b).

state is locally stable can cross into a region in which an interior steady state exists. However, this change is not path-dependent. The steady states are pinned down by the contemporaneous value of μ. If μ reverts to its original level and the parameter configuration again implies that the biased steady state is the unique steady state, the dynamics converge back to that steady state.

The absence of path dependence in the individual-bias channel helps rationalize a recurring empirical finding. Many contact interventions generate short-run improvements in attitudes that dissipate once the intervention ends (Scacco and Warren, 2018; Paluck et al., 2019, 2021). Temporary increases in μ alter the transition path but not the steady state unless they permanently change reinforcement. Bias reverts once interaction patterns return to baseline. Durable effects arise only when contact reshapes the underlying reinforcement environment itself (Lowe, 2021). The framework therefore provides an economic explanation for why exposure-based interventions frequently fail to produce lasting change absent sustained institutional or social reinforcement.

3.3 Societal Feedback

I now isolate the societal-feedback channel by shutting down effort and individual feedback. Can aggregate social feedback alone can generate persistent effects from increases in out-group contact? The answer now depends on the shape of $\theta(\bar{\pi})$. When $\theta(\bar{\pi})$ is linear or convex, the dynamics do not generate threshold-based path dependence from temporary changes in out-group contact. When $\theta(\bar{\pi})$ is sufficiently S-shaped, threshold effects can emerge, giving rise to multiple steady states and distinct basins of attraction.

Fix $\beta(b_i, b_j) = s(e_i, e_j) = 0$ for all e_i, e_j, b_i and b_j. In this case, the probability of a positive interaction is given by $\rho_{b_i,b_j}(e_i, e_j) = 1 - \theta(\bar{\pi})$, yielding the evolutionary dynamic

$$\Delta\pi_\tau = (1 - \eta_\tau)(\theta(\bar{\pi}) - \pi_\tau) + \eta_\tau(1 - \pi_\tau)\varepsilon. \tag{13}$$

Hence, steady states are given by the solutions to the system

$$\pi_\tau = \gamma(\eta_\tau, \varepsilon)\theta(\bar{\pi}) + [1 - \gamma(\eta_\tau, \varepsilon)] \times 1, \tag{14}$$

where

$$\gamma(\eta_\tau, \varepsilon) = \frac{1 - \eta_\tau}{1 - \eta_\tau(1 - \varepsilon)} \equiv \gamma_\tau \in (0, 1),$$

which is strictly decreasing in both η_τ and ε (and thus increasing in μ). Define $\bar{\gamma} = q\gamma_1 + (1 - q)\gamma_2$.

This system can be reduced by studying the aggregate prevailing bias $\bar{\pi}$. Using (14) to compute $\bar{\pi} = q\pi_1 + (1-q)\pi_2$ yields

$$\bar{\pi} = \bar{\gamma}\left(\theta(\bar{\pi}) - 1\right) + 1. \tag{15}$$

Thus, the steady states directly depend on the shape of $\theta(\bar{\pi})$. I consider three possibilities for θ: linear, convex, and S-shaped.[17] When linear, $\theta(\bar{\pi}) = \bar{\pi}$. When S-shaped, I assume

$$\lim_{\bar{\pi}\to 0^+} \frac{d\theta(\bar{\pi})}{d\bar{\pi}} < 1, \quad \lim_{\bar{\pi}\to 1^-} \frac{d\theta(\bar{\pi})}{d\bar{\pi}} < 1.$$

These assumptions are consistent with Section 2.1 for $k = 1$ (linear) and k large (S-shaped).

Proposition 4. *Fix $s(e_i, e_j) = 0$ and $\beta(b_i, b_j) = 0$ for all e_i, e_j, b_i, and b_j. The biased steady state exists. Moreover, if $\varepsilon = 0$ and $\mu > -1$, then the following statements hold.*

(i) *The unbiased steady state exists.*

(ii) *If $\theta(\bar{\pi})$ is convex, then almost all initial conditions converge to the unbiased steady state.*

(iii) *If $\theta(\bar{\pi})$ is linear, then initial conditions (π_1^0, π_2^0) converge to the steady state*

$$\pi_1^\infty = \pi_2^\infty = \frac{q^2}{q^2 + (1-q)^2}\pi_1^0 + \frac{(1-q)^2}{q^2 + (1-q)^2}\pi_2^0.$$

If $\varepsilon > 0$ and $\mu > -1$, then the following statements hold.

(iv) *If $\theta(\bar{\pi})$ is linear, then all initial conditions converge to the biased steady state.*

(v) *If $\theta(\bar{\pi})$ is convex and $\bar{\gamma}\frac{d\theta(1)}{d\bar{\pi}} > 1$, then in addition to the biased steady state, there is a unique interior steady state $\bar{\pi}^* \in (0, 1)$ to which almost all initial conditions converge. Otherwise, the biased steady state is the unique globally stable steady state.*

If $\mu = -1$, then $\pi_\tau^\infty = 1$ when $\varepsilon > 0$ and $\pi_\tau^\infty = \pi_\tau^0$ when $\varepsilon = 0$.

In line with the previous mechanisms, the steady-state is is independent of μ. However, the societal-feedback channel differs from the previous benchmarks because the steady states depend on the shape of aggregate feedback.[18] When θ is linear or convex, the dynamics do not generate threshold-based path dependence from temporary contact changes. Depending on the shape of θ and the

[17] The community norm enforcement foundation in Section 2.1 provides a concrete case, $k = 1$, $k = 2$, and $k \geq 3$, respectively, for these three shapes.

[18] Allowing θ to be concave does not substantially add anything to the results. When $\varepsilon = 0$, there is convergence to the biased steady state from almost all initial conditions and when $\varepsilon > 0$, the results are identical to the linear case. See the proof for details

value of ε, the system converges either to a unique steady state or to a steady state pinned down by initial conditions in the knife-edge linear case. The novelty arises when feedback is sufficiently S-shaped. In that case, small differences in aggregate bias can be amplified rather than dampened, generating tipping points and multiple steady states.

Proposition 5. *Fix $s(e_i, e_j) = 0$ for all e_i, e_j, $\beta(b_i, b_j) = 0$ for all b_i, b_j, and $\theta(\bar{\pi})$ is S-shaped. The biased steady state exists. If $\varepsilon = 0$ and $\mu > -1$, then there are three steady states: the unbiased steady state, the biased steady state, and an interior steady state $\bar{\pi}^*$.*

If $\varepsilon > 0$ and $\mu > -1$, then there exists a $\bar{x} \in (0, 1)$ such that there are three steady states: a stable interior steady state $\bar{\pi}^*$, an unstable interior tipping point $\bar{\pi}^*_U > \bar{\pi}^*$, and the biased steady state iff $\frac{d\theta(\bar{x})}{d\bar{\pi}} > \bar{\gamma}^{-1}$. Otherwise, the biased steady state is unique.

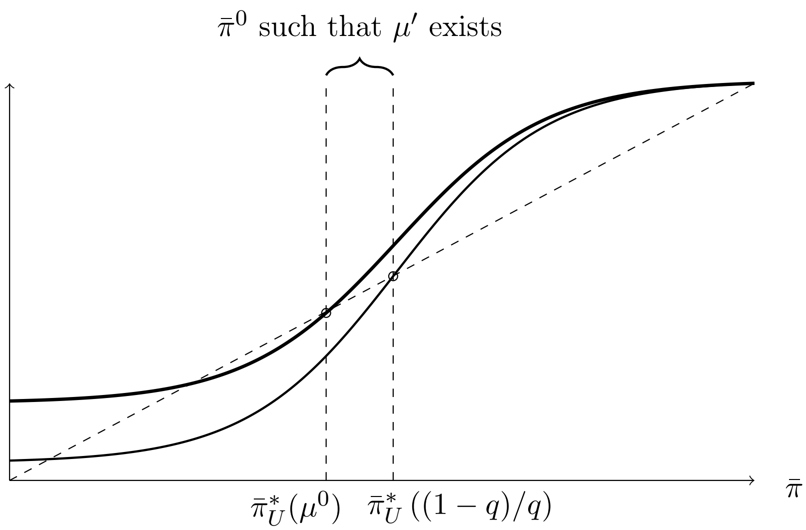
If $\mu = -1$, then $\pi^\infty_\tau = 1$ for all initial conditions when $\varepsilon > 0$ and $\pi^\infty_\tau = \pi^0_\tau$ when $\varepsilon = 0$.

Proposition 5 establishes that S-shaped societal feedback fundamentally changes the dynamics. When θ is S-shaped, three steady states coexist: a low-bias steady state, a high-bias steady state, and an unstable threshold separating their basins of attraction. Convergence depends on where the society begins relative to this tipping point. In this environment, out-group contact can alter which steady state prevails.

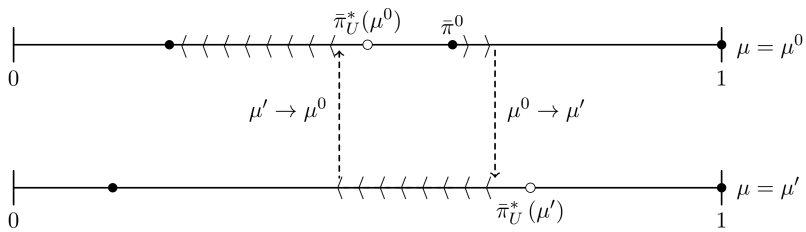
This observation raises a natural policy question. If the economy begins in the basin of attraction of the high-bias steady state, can a temporary increase in out-group exposure push aggregate bias below the tipping point into the low-bias basin of attraction and placing the economy on a permanently lower-bias path? Proposition 6 shows that, under the conditions of Proposition 5, the answer is yes. Let μ^0 denote the initial match parameter and let $\bar{\pi}^*_U(\mu)$ denote the tipping point given μ.

Proposition 6. *Fix $s(e_i, e_j) = 0$ and $\beta(b_i, b_j) = 0$ for all e_i, e_j, b_i, b_j. Suppose $\varepsilon > 0$, and $\theta(\bar{\pi})$ is S-shaped such that there are three steady states. If $\bar{\pi}^0 \in \left(\bar{\pi}^*_U(\mu^0), \bar{\pi}^*_U\left(\frac{1-q}{q}\right)\right)$, then there exists a $\mu' > \mu^0$ such that temporarily increasing μ^0 to μ' yields a persistent reduction in aggregate bias.*

Proposition 6 delivers the minimal mechanism through which temporary contact can generate persistent change. S-shaped societal feedback establishes a tipping point, and contact need only last long enough to move society across them. If initial aggregate bias lies just inside the basin of the



(a) Evolution of $\bar{\pi}$ under $\mu = \mu^0$ and $\mu = \frac{1-q}{q}$.



(b) Transition path given initial biases (π_1^0, π_2^0) and initial average bias $\bar{\pi}^0 = q\pi_1^0 + (1-q)\pi_2^0$ and temporary increase in μ from μ^0 to μ'.

Figure 2: Graphical illustration of Proposition 6.

high-bias steady state, raising μ expands the basin of attraction of the low-bias steady state, switching the basin in which current bias lies. Once bias falls sufficiently, reverting μ restores the original tipping point, but bias now lies on the low-bias side of it. The intervention need not be permanent; it need only last long enough to move the economy across the tipping point. Figure 2 illustrates this mechanism.

This mechanism rationalizes why some contact interventions produce durable change while others do not. Field evidence shows that structured, cooperative exposure can generate persistent reductions in prejudice (e.g., Bazzi et al., 2019; Lowe, 2021), whereas short-lived or unstructured contact often yields effects that attenuate once exposure ends (Scacco and Warren, 2018; Paluck et al., 2019). Why do some interventions succeed while others fail? The model provides a unifying interpretation. Contact has lasting effects only when it shifts the society across the relevant tipping point.

The analysis thus far isolates the minimal mechanism required for temporary contact to generate permanent change. In practice, however, effort, individual feedback, and societal feedback operate simultaneously. I now reintroduce all three channels and characterize the full dynamics of the model, showing how their interaction determines when contact reduces bias and when it instead reinforces high-bias outcomes.

4 Dynamics of the Full Model

I now reintroduce all three channels: effort, individual feedback, and societal feedback. The full model nests each benchmark in Section 3, but their interaction generates new forces absent in isolation. Effort counteracts in-group bias. As a result, the evolution of bias may be either globally convergent to a unique steady state or threshold-driven, depending on parameters. This section characterizes steady states of the full system and identifies when temporary increases in out-group contact can generate permanent change.

Lemma 1. *Conditional on individuals i and j matching with biases b_i and b_j, the probability of a positive interaction is monotonic in $\bar{\pi}$ iff*

$$V \leq \hat{V}(b_i, b_j) \equiv \frac{1}{1 - \beta(b_i, b_j)}.$$

Lemma 1 highlights the key interaction between effort, individual feedback, and societal feedback. In the societal-feedback benchmark of Section 3.3, higher aggregate bias mechanically reduced the probability of a positive interaction. Incorporating effort, individuals respond to this bias. When V is small, effort is weak and the probability of success declines monotonically in $\bar{\pi}$. When V is sufficiently large, effort responds strongly enough that increases in $\bar{\pi}$ induce additional investment, dampening the effect of aggregate bias on interaction outcomes and potentially rendering success probabilities non-monotonic in $\bar{\pi}$. High-value environments therefore attenuate the propagation of bias. This has a direct policy implication. V-based interventions may be most effective precisely in highly biased societies, where the incentive to invest in effort is greatest. This reverses the intuition from the benchmark, where high-bias environments are exactly the ones most resistant to intervention.

Effort also eliminates the fully biased steady state.

Lemma 2. *Suppose that $\mu > -1$. The biased steady state does not exist. Additionally, the unbiased steady state exists iff $\varepsilon = 0$.*

In the individual feedback and societal feedback benchmarks, the fully biased steady state was self-sustaining because no endogenous force reduced bias once it was universal. With effort, however, biased individuals still invest to secure positive interactions. As long as $\mu > -1$, out-group contact occurs with positive probability, and effort generates positive interactions that reduce bias. The biased steady state therefore disappears.

To characterize the multiplicity of steady states, define the *slope envelope* as

$$\tilde{p}'(\bar{\pi}) = \frac{d\theta(\bar{\pi})}{d\bar{\pi}} \times \max_{\beta\in[0,1]} \frac{(1-\beta)|1-(1-\beta)\theta(\bar{\pi})V|}{(1+(1-\beta)\theta(\bar{\pi})V)^3},$$

and set $\tilde{P}' := \sup_{\bar{\pi}\in[0,1]} \tilde{p}'(\bar{\pi})$, which characterizes the maximal steepness of the inflection point when societal feedback is self-reinforcing (S-shaped). Additionally, define

$$\bar{\Delta} = \sup_{\bar{\pi}\in[0,1]} \max_{\beta,\beta'\in\{0,\beta_1,\beta_0\}} |m_\beta(\bar{\pi}) - m_{\beta'}(\bar{\pi})| \geq 0,$$

where $m_\beta(\bar{\pi}) = 1 - \rho_{b_i,b_j}$ for $\beta = \beta(b_i, b_j)$.

The term $\bar{\Delta}$ is the largest gap in interaction failure probabilities across all levels of aggregate bias and bias configurations.

Proposition 7. *Suppose $\mu > -1$. The following statements hold.*

(i) *If $2\bar{\Delta} + \tilde{P}' < \gamma_2^{-1}$, then there is a unique steady state $(\pi_1^\infty, \pi_2^\infty) \in [0,1)^2$.*

(ii) *Suppose $\varepsilon > 0$ and $\frac{dm_{\beta_0}(\bar{\pi})}{d\bar{\pi}}$ is single-peaked on $(0,1)$ with $\bar{\pi}_M = \arg\max_{\bar{\pi}} \frac{dm_{\beta_0}(\bar{\pi})}{d\bar{\pi}} \in (0,1)$. There exists a $\hat{\Delta}$ such that if $\tilde{p}'(0)$, $\tilde{p}'(1) < \bar{\gamma}^{-1} < \frac{dm_{\beta_0}(\bar{\pi}_M)}{d\bar{\pi}}$ and $\bar{\Delta} < \hat{\Delta}$, then there are at least three interior steady states.*

Let $\bar{\pi}_1^*$, $\bar{\pi}_2^*$, and $\bar{\pi}_3^*$ denote the steady state aggregate bias levels. Here, $\bar{\pi}_2^*$ is the tipping point, analogous to $\bar{\pi}_U^*$ from Section 3.3. Because $\tilde{P}'$ is decreasing in V, uniqueness in statement (i) is more easily satisfied when V is large, individuals then respond aggressively to rising bias, dampening the aggregate feedback required for multiplicity. Effort therefore attenuates the nonlinearities necessary for tipping dynamics. Invoking the community norm enforcement interpretation of $\theta(\bar{\pi})$ from Section 2.1, for any $V > 0$ and maintaining $\varepsilon >$ and $\bar{\Delta} < \hat{\Delta}$, there always exists k sufficiently large to satisfy the slope conditions of (ii). The latter condition is least restrictive precisely when tipping dynamics are most pronounced (see footnote 25 in the Appendix).

Let $\Omega = (\theta, \beta_0, V, q, \mu, \varepsilon)$ denote the set of parameters under which the full model admits three steady states and let Ω_S denote the analogous set under the societal feedback benchmark. Similarly, let $\tilde{\Omega}$ and $\tilde{\Omega}_S$ denote the respective sets such that a temporary increase in μ can generate a persistent reduction in long run bias under the full model and societal-feedback benchmark.

Proposition 8. *Suppose $\varepsilon > 0$ and parameters are contained in Ω for all $\mu \in \left[\mu^0, \frac{1-q}{q}\right]$. If $\bar{\pi}^0 \in \left(\bar{\pi}_2^*(\mu^0), \bar{\pi}_2^*\left(\frac{1-q}{q}\right)\right)$, then there exists a $\mu' > \mu^0$ such that temporarily increasing μ^0 to μ' yields a persistent reduction in long-run bias with convergence to $\bar{\pi}_1^*(\mu^0)$.*

Proposition 8 shows that basin-switching remains possible in the full model. However, because individuals increase effort when $\bar{\pi}$ rises, the feedback from aggregate bias to contact outcomes is dampened. Effort therefore shrinks the parameter region supporting multiplicity, so the set of environments in which temporary increases in contact can permanently shift the society across basins of attraction is smaller than in the societal-feedback benchmark.

Proposition 9. *Suppose $\mu > -1$ and $\varepsilon > 0$.*

(i) *For any $V > 0$ and $\beta_0 \in [0, 1)$, $\Omega \subset \Omega_S$.*

(ii) *$\tilde{\Omega} \subseteq \tilde{\Omega}_S$, strictly so for $V > 0$ and $\beta_0 \in [0, 1)$.*

Taken together, the full model reveals a central tradeoff introduced by effort. Higher-value contact reduces steady-state in-group bias by inducing preventive investment in effort, thereby stabilizing the dynamics. At the same time, this stabilization weakens the sensitivity of contact outcomes to aggregate bias. Because individuals respond to rising $\bar{\pi}$ by increasing effort, the societal feedback loop is dampened and the nonlinearity required for tipping dynamics is harder to sustain. High-value contact environments therefore lower steady-state in-group bias while simultaneously limiting the scope for temporary exposure to generate permanent regime shifts. In this sense, effort changes both the level and the shape of the dynamics. It lowers long-run bias, but it also shrinks the set of environments in which μ-based policy succeeds.

Proposition 9 establishes that the parameter region admitting μ-based tipping is strictly smaller in the full model than in the benchmark. If the economy is deeply entrenched in high bias, so that

$\bar{\pi}^0$ lies above the window $\left(\bar{\pi}_2^*(\mu^0), \bar{\pi}_2^*\left(\frac{1-q}{q}\right)\right)$, it cannot be rescued by any temporary increase in μ. The following proposition shows that, under the maintained assumptions, increasing V provides an alternative route to permanent change that does not require the initial condition to lie inside the μ-window. Let $\bar{\pi}_\ell^*(V)$ denote the steady states as functions of V.

Proposition 10. *Suppose $\varepsilon > 0$, $\mu > -1$, and parameters are contained in Ω.*

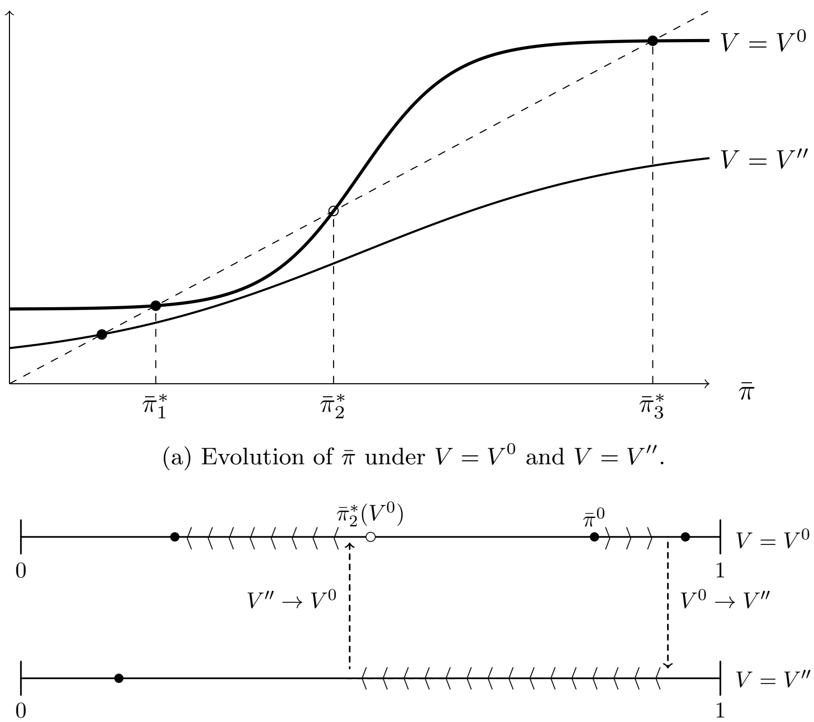
(i) *There exists $V' > V^0$ such that for all $V \geq V'$, the full model admits a unique globally attracting steady state $\bar{\pi}_1^*(V) \in (0,1)$, with $\bar{\pi}_1^*(V) < \bar{\pi}_2^*(V^0)$.*

(ii) *For any $\bar{\pi}^0 \in \left(\bar{\pi}_2^*(V^0), 1\right)$, temporarily increasing V from V^0 to any $V'' > V'$ satisfying (i) yields a persistent reduction in long-run bias, with the economy converging to $\bar{\pi}_1^*(V^0)$.*

Proposition 10 complements Proposition 8 but works through a fundamentally different mechanism. A temporary increase in μ works by shifting the tipping point $\bar{\pi}_2^*(\mu)$ upward, expanding the basin of attraction of the low-bias steady state until the current state falls inside it; this requires $\bar{\pi}^0$ to lie within a specific window of initial conditions. A temporary increase in V works by raising the return to effort. Individuals then invest heavily in positive contact even in high-bias environments, eroding the self-reinforcing dynamic that sustains the high-bias trap and leaving a unique low-bias steady state. During the intervention, aggregate bias reduces towards the unique low-bias steady state. The restriction on initial conditions near the tipping point disappears entirely. Figure 3 illustrates this mechanism.

The two instruments therefore differ in robustness. μ-based interventions are sufficient only when the economy is already near the tipping point, whereas V-based interventions admit no such constraint. Proposition 10 shows that a sufficiently strong V-intervention generates permanent change from any initial condition (provided that aggregate bias was trending towards the high-bias steady state), while Proposition 8 shows that a μ-intervention succeeds only when prevailing bias lies within a window around the tipping point. This asymmetry implies that the two instruments are complements rather than substitutes. A temporary V-intervention can move a deeply entrenched economy into the window from which a subsequent μ-intervention completes the transition to the low-bias steady state, allowing the V-intervention to be withdrawn earlier than under a pure V-policy. Formalizing this sequencing logic requires two preliminary results.

The first derives a bound on how fast aggregate bias can fall in a single period under the baseline



(a) Evolution of $\bar{\pi}$ under $V = V^0$ and $V = V''$.

(b) Transition path given initial biases (π_1^0, π_2^0) and initial average bias $\bar{\pi}^0 = q\pi_1^0 + (1-q)\pi_2^0$ and temporary increase in V from V^0 to V''.

Figure 3: Graphical illustration of Proposition 10.

matching structure.

Lemma 3. *For any $\mu > -1$, $\bar{\pi}^t - \bar{\pi}^{t+1} \leq 2q(1-q)(1+\mu) \equiv C^*(\mu, q)$ for all $t \geq 0$.*

The second shows that the unstable tipping point moves monotonically in both policy instruments.

Corollary 3. *Under Ω with $\mu > -1$ and $\varepsilon > 0$, the unstable tipping point $\bar{\pi}_2^*(\mu, V)$ is strictly increasing in both μ and V.*

Fix a candidate sequential matching level $\tilde{\mu} \in \left(\mu^0, \frac{1-q}{q}\right]$ and a V-intervention level V'' satisfying Proposition 10(i). Define crossing times along the trajectory induced by the constant policy (V'', μ^0):

$$T_V \equiv \min\left\{t \geq 0 : \bar{\pi}^t < \bar{\pi}_2^*(\mu^0, V^0)\right\},$$

$$\tilde{T}_V \equiv \min\left\{t \geq 0 : \bar{\pi}^t < \bar{\pi}_2^*(\tilde{\mu}, V^0)\right\},$$

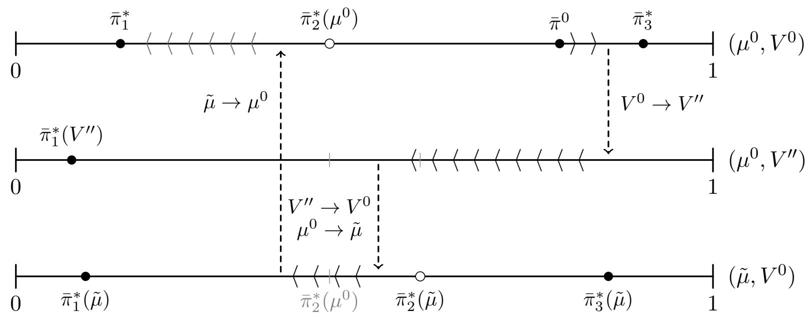


Figure 4: Transition path given initial biases (π_1^0, π_2^0) and initial average bias $\bar{\pi}^0 = q\pi_1^0 + (1-q)\pi_2^0$ and temporary increase in V from V^0 to V'', followed by returning V'' to V^0 and instituting a temporary increase in μ from μ^0 to $\tilde{\mu}$.

and the μ-phase duration along the trajectory induced by $(\tilde{\mu}, V^0)$ initiated at $\bar{\pi}^{\tilde{T}_V}$:

$$T_\mu \equiv \min\left\{t \geq 0 : \bar{\pi}^{\tilde{T}_V + t} < \bar{\pi}_2^*(\mu^0, V^0)\right\}.$$

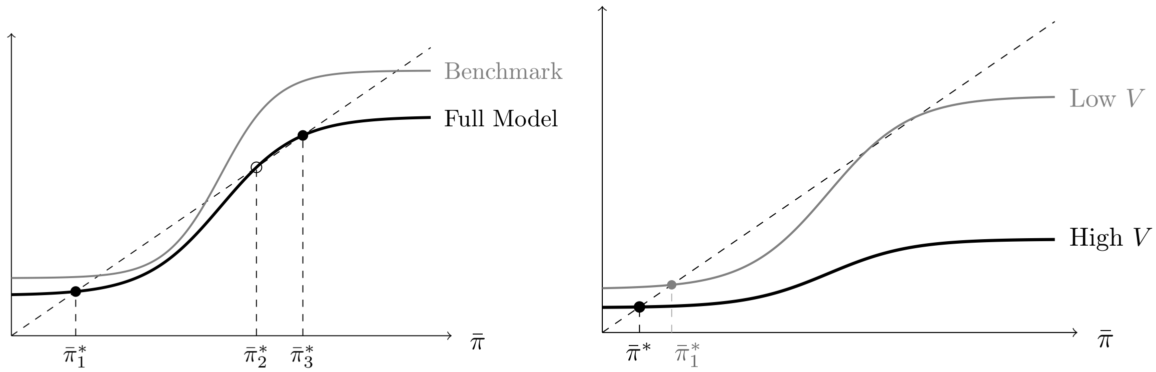
To obtain a nontrivial sequencing result, assume that a μ-intervention alone fails; i.e., for every feasible $\mu' \in \left(\mu^0, \frac{1-q}{q}\right]$, $\bar{\pi}^0 \geq \bar{\pi}_2^*(\mu', V^0)$. Additionally, assume that there exists $\tilde{\mu} \in \left(\mu^0, \frac{1-q}{q}\right]$ such that $C^*(\mu^0, q) < \bar{\pi}_2^*(\tilde{\mu}, V^0) - \bar{\pi}_2^*(\mu^0, V^0)$.

Proposition 11. *Suppose $\varepsilon > 0$, $\mu^0 > -1$, and parameters are contained in Ω for all $\mu \in \left[\mu^0, \frac{1-q}{q}\right]$, with stable aggregate bias levels $\bar{\pi}_1^*(\mu, V) < \bar{\pi}_3^*(\mu, V)$ and unstable tipping point $\bar{\pi}_2^*(\mu, V)$. Fix (μ^0, V^0) and $\bar{\pi}^0 \in \left(\bar{\pi}_2^*(\mu^0, V^0), 1\right)$. Then $T_V, \tilde{T}_V, T_\mu < \infty$ and the combined policy that raises V^0 to V'' on $t \in [0, \tilde{T}_V)$, μ^0 to $\tilde{\mu}$ on $t \in [\tilde{T}_V, \tilde{T}_V + T_\mu)$, then ends the intervention yields a persistent reduction in steady state aggregate bias. Additionally, $\tilde{T}_V < T_V$, so the V-intervention is withdrawn strictly earlier under sequencing than under the V-only policy.*

Corollary 4. *Under the conditions of Proposition 11, let $c_V > c_\mu > 0$ denote the per-period costs of the V- and μ-interventions, respectively. The sequencing policy reduces the total intervention cost relative to the pure-V policy iff*

$$\frac{c_V}{c_\mu} > \frac{T_\mu}{T_V - \tilde{T}_V}.$$

Figure 4 illustrates the transition path of the sequenced policy. Proposition 11 shows that sequencing can shorten the required V-phase, though not necessarily the total intervention duration. A μ-intervention need only last long enough to push the trajectory below the baseline tipping point



(a) Full model low V v. benchmark.

(b) Full model, low V v. high V.

Figure 5: Endogenous effort attenuates the societal feedback sustaining tipping dynamics (Panel (a), Proposition 9). As V increases, this attenuation eliminates the high-bias steady state entirely (Panel (b), Proposition 10)

$\bar{\pi}_2^*(\mu^0, V^0)$, whereas a pure V-intervention must do so without the aid of a temporary increase in μ. Whether sequencing reduces total duration depends on the speed of the μ-induced dynamics relative to the remaining V-phase $T_V - \tilde{T}_V$. A more aggressive $\tilde{\mu}$ widens the gap $\bar{\pi}_2^*(\tilde{\mu}, V^0) - \bar{\pi}_2^*(\mu^0, V^0)$, relaxing the assumption on $C^*(\mu^0, q)$ and shortening $\tilde{T}_V$, but may lengthen T_μ if the state enters the tipping window close to its upper boundary. The choice of $\tilde{\mu}$ therefore trades off the duration of the V-phase against that of the μ-phase. However, Corollary 4 shows that when value-enhancing interventions are sufficiently costly relative to temporary changes in contact, sequencing can reduce total weighted intervention cost by shortening the required V-phase.

The choice of $\tilde{\mu}$ involves a genuine tradeoff. By Corollary 3, $\bar{\pi}_2^*(\tilde{\mu}, V^0)$ is strictly increasing in $\tilde{\mu}$, so a more aggressive matching mandate shortens the V-phase by raising the threshold the trajectory must cross before the μ-phase begins. However, its effect on the μ-phase duration T_μ is ambiguous. A higher $\tilde{\mu}$ speeds convergence once imposed, but it also begins from a state with higher prevailing bias. The preferred value of $\tilde{\mu}$ therefore need not equal the maximum feasible matching rate $(1-q)/q$.

Sections 3 and 4 together clarify the conditions under which contact alters the steady-state in-group bias. When feedback is linear or convex, exposure affects long-run bias but does not generate threshold-driven regime shifts. Figure 5 illustrates the two central mechanisms. Endogenous effort shrinks the parameter region in which μ-based tipping is possible, but simultaneously creates the conditions under which V-based regime elimination becomes effective. The contact hypothesis therefore holds when social feedback is S-shaped, and the instrument best suited to exploiting that feedback depends on where the population stands relative to the tipping point.

5 Policy Implications

The model's three main results each carry a distinct policy implication, developed in turn below.

5.1 Effort and the Value of Successful Interaction

Policies that increase V correspond to institutional interventions that increase the returns to successful interaction. This includes team-based pay that ties compensation to cooperative output, school environments that create shared goals across group lines, or professional settings structured around joint production. A sufficiently large temporary increase in V eliminates the high-bias steady state for the duration of the intervention, driving the economy toward a unique low-bias steady state from any initial condition (Proposition 10).

This prediction finds support in studies of integration into high-value settings (Rao, 2019; Boisjoly et al., 2006; Beaman et al., 2009; Hjort, 2014). The model's mechanism implies a sharper reading of these findings than a directional prediction alone. Because effort both reduces steady-state bias and attenuates the societal feedback sustaining the tipping point, environments with higher interaction stakes should exhibit lower steady-state in-group bias but a narrower region in which temporary contact interventions generate lasting change. Consistent with this, Rao (2019) finds that the largest reductions in discriminatory behavior arise specifically in economic exchange tasks, which are precisely the interactions where V is highest and, by the model's logic, where effort is most strongly induced. A further practical constraint is that if institutional integration is perceived as culturally threatening, individuals may strategically withdraw, endogenously recreating assortative sorting and offsetting the gains from increasing V (Carvalho et al., 2017; Carvalho and Sacks, 2024; Carvalho et al., 2024b).

5.2 Assortative and Disassortative Matching

Policies that raise μ toward disassortative matching correspond to integration mandates, diversity requirements, school assignment policies, and workplace composition rules that increase the frequency of out-group interaction. Conditional on the economy beginning in the basin of attraction of the high-bias steady state, a temporary increase in μ generates a persistent reduction in bias if and only if prevailing average bias lies between the tipping point under the current contact rate and the tipping point under the maximum feasible contact rate.[19]

[19] Formally, the condition is $\bar{\pi}^0 \in \big(\bar{\pi}_2^*(\mu^0), \bar{\pi}_2^*\big(\frac{1-q}{q}\big)\big)$, where $\frac{1-q}{q}$ is the upper bound on μ imposed by market clearing, $q(1-\eta_1) = (1-q)(1-\eta_2)$.

This window-dependence has a direct empirical counterpart. The tipping literature in urban economics documents that neighborhoods and schools exhibit sharp threshold dynamics in racial composition. Once the share of one group crosses a critical level, the social environment transitions rapidly and persistently to a segregated state (Schelling, 1971; Card et al., 2008). In the model, these are the kinds of environments where in-group bias feeds back strongly enough on interaction outcomes to support multiple steady states, and where μ-based tipping is feasible.[20]

The window condition offers a specific, falsifiable account of cross-setting heterogeneity in contact interventions. Lowe (2021) finds persistent reductions in discrimination following a cooperative cricket program in India, while Mousa (2020) finds that a structurally similar intervention in post-ISIS Iraq produces effects confined to the contact setting. Consistent with the window interpretation, Ghosh et al. (2026) find that a structured youth camp intervention in West Bengal, India generates persistent reductions in in-group bias one year after the camps ended. The model identifies three distinct channels through which such divergence can arise. First, the two populations may have differed in their position relative to the tipping point, with prevailing bias in one setting lying within $\left(\bar{\pi}_2^*(\mu^0), \bar{\pi}_2^*\left(\frac{1-q}{q}\right)\right)$ and in the other lying outside it. Second, the settings may have differed in whether societal feedback is sufficiently S-shaped to generate tipping dynamics at all, so that the same initial bias level produces a unique stable steady state in one environment and multiple steady states in the other. Third, even when both settings exhibit tipping dynamics, differences in the shape of θ, for instance the size of the relevant social reference group in the community norm enforcement microfoundation, shift the location and width of the tipping window itself, so that identical initial bias levels fall inside the window in one context and outside it in the other. All three channels generate predictions absent from heterogeneity-based explanations: whether a contact program succeeds depends on measurable features of the societal environment, not merely on implementation details.

Steinmayr (2021) finds that communities exposed to refugee resettlement exhibit reduced support for far-right parties when meaningful direct contact occurs, but increased support when exposure occurs without genuine interaction. The model offers a specific account of this asymmetry. Increasing μ raises the rate of out-group contact but does not directly raise the quality of interactions; without a corresponding V, the model predicts that exposure alone is insufficient to cross the tipping threshold,

[20] Interventions that successfully pushed neighborhoods or schools across such thresholds, as in sustained court-ordered desegregation, generated lasting integration (Guryan, 2004); interventions that failed to sustain the mandate were reversed once legal pressure lapsed, with corresponding deterioration in outcomes (Lutz, 2011; Billings et al., 2014).

leaving steady-state bias unchanged or inducing reversion. Meaningful contact, by contrast, engages the effort channel and can generate the positive interaction outcomes through which bias is reduced. The model does not account for backlash of the kind documented by Enos (2014), where out-group contact actively increases exclusionary attitudes; that pattern likely reflects mechanisms outside the model's scope, such as identity threat or status competition. Within interventions that do succeed, however, the model predicts that effects should be asymmetric across majority and minority participants. Ghosh et al. (2026) find, in a randomized youth camp experiment in West Bengal, India, that higher out-group contact intensity backfires among Hindu majority participants but not Muslim minority participants, consistent with the model's prediction that majority members, facing higher baseline in-group exposure, are more adversely affected by contact interventions that fall short of the tipping threshold.

The model also implies that contact policy changes the cross-group distribution of bias, not just its average level. Corollary 2 establishes that, under empirically plausible conditions, the majority is strictly more biased than the minority in any interior steady state, reflecting the majority's higher in-group interaction rate and greater accumulated in-group reinforcement.[21] A disassortative matching policy reduces in-group exposure for both groups, but the reduction is proportionally larger for the majority. A uniform increase in μ therefore narrows the bias gap between groups, generating convergence in the distribution of bias alongside a reduction in average bias. This is consistent with evidence that contact effects are often stronger among majority than minority group members (Tropp and Pettigrew, 2005).

6 Concluding Remarks

This paper develops a dynamic model of social interaction to explain why some temporary interventions durably reduce bias while others fail. The key insight is that permanent change requires self-reinforcing societal feedback strong enough to generate a tipping point. When that condition holds, temporarily increasing the rate of out-group contact reduces long-run bias only if prevailing bias lies sufficiently close to the tipping point; otherwise, bias reverts once the intervention ends. Raising the value of successful interaction operates over a broader range of initial conditions, but does so through the same effort response that also weakens the multiplicity needed for temporary contact alone to generate lasting change. Finally, under empirically plausible conditions,

[21] The conditions are $\mu > -1$, so that out-group contact occurs with positive probability, and $\varepsilon > 0$, so that in-group interactions generate positive reinforcement (ε can be arbitrarily small).

the model predicts a majority–minority asymmetry in long-run bias, with majority members more biased than minority members in any interior steady state because differential in-group exposure generates asymmetric reinforcement.

Together, these results provide a threshold-based framework for understanding the heterogeneous outcomes of contact-based interventions documented in the empirical literature (Paluck et al., 2021; Mousa, 2020; Lowe, 2021, 2025). When initial bias is too entrenched for contact alone to succeed, the framework implies a sequencing logic, where interventions that raise the value of successful interaction can first move the population into the region where a temporary increase in out-group contact can produce lasting change. The formal sequencing result further shows that such a policy sequence can shorten the duration of the value-enhancing intervention relative to a pure-V policy, though not necessarily the total duration of intervention.

Having isolated the core mechanism in a tractable setting, the framework suggests three directions for future research. First, the model yields directly testable comparative predictions. In particular, environments with higher stakes for successful interaction should exhibit lower steady-state bias but a smaller region in which temporary contact interventions generate lasting change, while sequenced interventions should outperform contact-only interventions in more deeply entrenched settings. Second, the model takes in-group bias as binary within types. A natural extension is to allow for a distribution of bias levels within each group, which could generate richer distributional dynamics, including whether contact reduces average bias across groups while increasing within-group polarization. Third, endogenizing the contact rate μ is a natural extension. When societal bias is high, individuals may prefer in-group interactions, reducing μ endogenously and reinforcing the aggregate bias that generated the preference for segregation in the first place (Currarini et al., 2009; Gentzkow and Shapiro, 2011; Schelling, 1971). This self-segregation channel would introduce a second feedback loop into the dynamics, potentially amplifying tipping behavior and shrinking the parameter region in which contact-based interventions succeed.[22]

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[22]Carvalho et al. (2025) shows that policies targeting a single identity dimension almost never achieve representative outcomes across all dimensions simultaneously, suggesting that matching mandates aimed at one dimension of identity may generate unintended consequences along others.

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Appendix

Proof of Proposition 1.

Proof. Conditional on i and j being matched with type-bias pairs (τ_i, b_i) and (τ_j, b_j), i chooses effort

$$e_i^*(b_i, b_j) = \arg\max_{e_i} \left(s(e_i, e_j) + (1 - s(e_i, e_j))[\beta(b_i, b_j) + (1 - \beta(b_i, b_j))(1 - \theta(\bar{\pi}))]V - \frac{1}{2}e_i^2 \right),$$

where $s(e_i, e_j) = e_i + e_j - e_i e_j$. Each j performs an analogous optimization. The corresponding first-order condition, evaluated at $e_i = e_i^*$ and $e_j = e_j^*$, is

$$e_i^* = (1 - e_j^*)(1 - \beta(b_i, b_j))\theta(\bar{\pi})V \iff \frac{e_i^*}{1 - e_j^*} = (1 - \beta(b_i, b_j))\theta(\bar{\pi})V.$$

As $\beta(b_i, b_j) = \beta(b_j, b_i)$, $e_i^* = e_j^*$. Setting $e_i^* = e_j^* = e^*$, substituting, and solving for e^* yields the expression in the proposition. The comparative statics follow immediately. $\square$

Proof of Proposition 2.

Proof. By Proposition 1, applying $\beta(b_i, b_j) = 0$ for all b_i, b_j and $\theta(\bar{\pi}) = 1$ for all $\bar{\pi}$ to (5) yields efforts $e_i^*(b_i, b_j) = e_j^*(b_i, b_j) = \frac{V}{1+V} \equiv e^*$. The expression for s^* follows.

As the probability of a positive interaction is dictated solely by effort when matched with an individual of a different type and randomness (via ε) when matched with an individual of the same type,

$$\Delta\pi_\tau(\pi_\tau) = (1 - \eta_\tau)\left[(1 - \pi_\tau)(1 - s^*) - \pi_\tau s^*\right] + \eta_\tau(1 - \pi_\tau)\varepsilon.$$

Observe that this expression is independent of $\pi_{-\tau}$ and linear in π_τ with

$$\Delta\pi_\tau(1) = -(1 - \eta_\tau)s^* < 0 < (1 - \eta_\tau)(1 - s^*) + \eta_\tau\varepsilon = \Delta\pi_\tau(0).$$

Hence, there is global convergence to a unique interior steady state π_τ^∞ from all π^0 given by the solution to

$$(1 - \eta_\tau)\left[(1 - \pi_\tau^\infty)(1 - s^*) - \pi_\tau^\infty s^*\right] + \eta_\tau(1 - \pi_\tau^\infty)\varepsilon = (1 - \eta_\tau)\left[1 - s^* - \pi_\tau^\infty\right] + \eta_\tau(1 - \pi_\tau^\infty)\varepsilon = 0. \quad (16)$$

Observe that if $\varepsilon = 0$ and $\eta_\tau = 1$, then $\Delta\pi_\tau(\pi_\tau) = 0$ for all π_τ, which implies that $\pi_\tau^\infty = \pi_\tau^0$. For all other parameter values, solving (16) for π_τ^∞ yields

$$\pi_\tau^\infty = 1 - \frac{(1 - \eta_\tau)s^*}{1 - \eta_\tau(1 - \varepsilon)} \in (0, 1],$$

proving statement (i).

To prove statement (ii), I compare π_1^∞ to π_2^∞:

$$\pi_1^\infty > \pi_2^\infty \iff \frac{(1 - \eta_2)}{1 - \eta_2(1 - \varepsilon)} < \frac{(1 - \eta_1)}{1 - \eta_1(1 - \varepsilon)}.$$

If $\varepsilon = 0$, then the above immediately collapses to an equality. Otherwise, it reduces to $\eta_1 > \eta_2$. Employing the definition of $\eta_\tau = \lambda(I_\tau) - \mu(1 - \lambda(I_\tau))$, $\eta_1 > \eta_2 \iff \mu > -1$. Hence, for all $\eta_1 > \eta_2$, or equivalently, for all $\mu > -1$, $\pi_1^\infty > \pi_2^\infty$ and at $\mu = -1$ and $\varepsilon > 0$, $\pi_1^\infty = \pi_2^\infty$. The statement $\pi_\tau^\infty = \pi_\tau^0$ when $\varepsilon = 0$ and $\mu = -1$ follows directly from (8), proving statement (ii).

Statement (iii) follows from differentiating π_τ^∞, given above, with respect to μ and V. As $\frac{d\eta_\tau}{d\mu} < 0$, $\frac{\partial\pi_\tau^\infty}{\partial\mu} < 0$. At $\mu = -1$, $\eta_\tau = 1$ and $\pi_\tau^\infty = 1$ for all V, so a strict decrease in V requires $\mu > -1$. $\square$

The following two Lemmas are useful for proving Proposition 3

Lemma 4. *Suppose that $s(e_i, e_j) = 0$ for all e_i, e_j, $\theta(\bar{\pi}) = 1$ for all $\bar{\pi}$, $\varepsilon = 0$, and $\mu > -1$. If $(\pi_1^\infty, \pi_2^\infty)$ solves $\Delta\pi_\tau = 0$ for $\tau = 1, 2$, then $\pi_1^\infty = \pi_2^\infty$.*

Proof. Set $s(e_i, e_j) = 0$ for all e_i, e_j and $\theta(\bar{\pi}) = 1$ for all $\bar{\pi}$. Then, by (9),

$$\Delta\pi_\tau = (1 - \eta_\tau)\Big[(1 - \pi_\tau)\big((1 - \pi_{-\tau})(1 - \beta_0) + \pi_{-\tau}(1 - \beta_1)\big) - \pi_\tau(1 - \pi_{-\tau})\beta_1\Big], \tag{17}$$

If $\mu > -1$, then $\eta_1, \eta_2 \in (0, 1)$, so $\Delta\pi_\tau = 0$ if and only if

$$\underbrace{(1 - \pi_\tau)\big((1 - \pi_{-\tau})(1 - \beta_0) + \pi_{-\tau}(1 - \beta_1)\big) - \pi_\tau(1 - \pi_{-\tau})\beta_1}_{\equiv F_\tau(\pi_\tau, \pi_{-\tau})} = 0$$

for $\tau = 1, 2$, which is equivalent to $F_1(\pi_1, \pi_2) = 0$ and $F_2(\pi_2, \pi_1) = 0$. Subtracting $F_2(\pi_2, \pi_1)$ from $F_1(\pi_1, \pi_2)$ yields $F_1(\pi_1, \pi_2) - F_2(\pi_2, \pi_1) = \pi_2 - \pi_1$. Thus, $F_1(\pi_1^\infty, \pi_2^\infty) = F_2(\pi_2^\infty, \pi_1^\infty) = 0$ implies $\pi_1^\infty = \pi_2^\infty$. □

Lemma 5. *Suppose that $s(e_i, e_j) = 0$ for all e_i, e_j, $\theta(\bar{\pi}) = 1$ for all $\bar{\pi}$, $\varepsilon = 0$, $\mu > -1$, $\beta_0 = 1$, and $\beta_1 = \frac{1}{2}$. Given any initial shares π_1^0 and π_2^0, there is convergence to $\pi_1^\infty = \pi_2^\infty = q\pi_1^0 + (1 - q)\pi_2^0$.*

Proof. Given the conditions set forth in the lemma, the law of motion reduces to

$$\begin{pmatrix} \pi_1^{t+1} \\ \pi_2^{t+1} \end{pmatrix} = \underbrace{\begin{pmatrix} 1 - \frac{(1-q)(1+\mu)}{2} & \frac{(1-q)(1+\mu)}{2} \\ \frac{q(1+\mu)}{2} & 1 - \frac{q(1+\mu)}{2} \end{pmatrix}}_{\equiv \mathbf{Q}} \begin{pmatrix} \pi_1^t \\ \pi_2^t \end{pmatrix},$$

with eigenvalues 1 and $\frac{1-\mu}{2} \in [0, 1)$ for $\mu > -1$. A direct calculation confirms that $q\pi_1^{t+1} + (1 - q)\pi_2^{t+1} = q\pi_1^t + (1 - q)\pi_2^t$, so the weighted average $q\pi_1^t + (1 - q)\pi_2^t$ is constant at $q\pi_1^0 + (1 - q)\pi_2^0$. As the second eigenvalue lies in $[0, 1)$, the solution converges to a multiple of the unit eigenvector $(1, 1)^T$, so $\pi_1^t - \pi_2^t \to 0$. Taking $t \to \infty$ with $\pi_1^\infty = \pi_2^\infty \equiv \pi^\infty$ and using the preserved quantity gives $(q+(1-q))\pi^\infty = q\pi_1^0+(1-q)\pi_2^0$, hence $\pi_1^\infty = \pi_2^\infty = q\pi_1^0 + (1 - q)\pi_2^0$. □

Proof of Proposition 3.

Proof. Set $s(e_i, e_j) = 0$ for all e_i, e_j, $\theta(\bar{\pi}) = 1$ for all $\bar{\pi}$, and $\varepsilon = 0$. First, observe that evaluating (9) at $(\pi_1, \pi_2) = (1, 1)$ yields $\Delta\pi_\tau = 0$ for each τ, so the biased steady state always exists.

To prove statements (i)–(iii), suppose $\varepsilon = 0$ and $\mu > -1$. By Lemma 4, any steady state is symmetric, so $\pi_1^\infty = \pi_2^\infty = \pi^*$ and the steady states are characterized by

$$F(\pi^*) = 0 \Longleftrightarrow (1 - \pi^*)[1 - \beta_0 + \pi^*(\beta_0 - 2\beta_1)] = 0. \tag{18}$$

From (18), the bracketed factor of $F(\pi)$ is linear in π, equaling $1 - \beta_0 \geq 0$ at $\pi = 0$ and $1 - 2\beta_1$ at $\pi = 1$. If $\beta_1 < \frac{1}{2}$, the bracket is positive on all of $[0, 1)$, so $F > 0$ and the biased steady state is globally attracting, proving statement (i) (the knife-edge $\beta_0 = 1$, $\beta_1 = \frac{1}{2}$ is Lemma 5). If $\beta_0 = 1$, then $F(\pi) = \pi(1-\pi)(1-2\beta_1)$, so the unbiased steady state exists, and $F < 0$ on $(0, 1)$ whenever $\beta_1 > \frac{1}{2}$, proving statement (ii). If $\beta_0 < 1$ and $\beta_1 > \frac{1}{2}$, the unique interior root $\pi^* = \frac{1-\beta_0}{2\beta_1-\beta_0} \in (0, 1)$ is the unique interior steady state, and the sign change of F at π^* together with monotonicity of the joint map implies global convergence from almost all initial conditions, proving statement (iii).

To prove statements (iv) and (v), suppose that $\varepsilon > 0$ and $\mu > -1$. Steady states solve the fixed-point system $\pi_1 = F_1(\pi_2; \varepsilon)$ and $\pi_2 = F_2(\pi_1; \varepsilon)$, where $F_\tau(\cdot; \varepsilon)$ is given by (10). Note that $F_\tau(\cdot; \varepsilon)$ is continuous and strictly increasing on $[0,1]$ since $\frac{\partial F_\tau(\pi_{-\tau}; \varepsilon)}{\partial \pi_{-\tau}} > 0$ for all $\pi_{-\tau} \in [0,1]$ by (11). Moreover, the biased steady state $(\pi_1^\infty, \pi_2^\infty) = (1,1)$ is always a steady state because $F_\tau(1; \varepsilon) = 1$ for each τ.

Define the composite map $H(x; \varepsilon) \equiv F_1(F_2(x; \varepsilon); \varepsilon)$ and $g(x; \varepsilon) \equiv H(x; \varepsilon) - x$. Then $(\pi_1^\infty, \pi_2^\infty)$ is a steady state iff π_1^∞ solves $g(\pi_1^\infty; \varepsilon) = 0$ and $\pi_2^\infty = F_2(\pi_1^\infty; \varepsilon)$. Differentiating H and evaluating at $x = 1$ yields

$$\frac{dH(1; \varepsilon)}{dx} = \frac{\partial F_1(1; \varepsilon)}{\partial \pi_2} \times \frac{\partial F_2(1; \varepsilon)}{\partial \pi_1}.$$

Suppose that $\frac{dH(1;\varepsilon)}{dx} \leq 1$. As $F_\tau(\cdot; \varepsilon)$ is strictly increasing, $H(\cdot; \varepsilon)$ is strictly convex on $[0,1]$, so $\frac{dH(x;\varepsilon)}{dx} \leq \frac{dH(1;\varepsilon)}{dx} \leq 1$ for all $x \in [0,1)$. As $\varepsilon > 0$ implies $H(0; \varepsilon) > 0$, we have $g(0; \varepsilon) > 0$, and combined with $g'(x; \varepsilon) < 0$ on $[0,1)$, it follows that $g(x; \varepsilon) > 0$ on $[0,1)$, so no interior steady state exists, proving statement (iv).

Now, suppose that $\frac{dH(1;\varepsilon)}{dx} > 1$. Then, $\frac{dg(1;\varepsilon)}{dx} > 0$, so for some small $\delta > 0$, $g(1-\delta; \varepsilon) = H(1-\delta; \varepsilon) - (1-\delta) < 0$. However, $\varepsilon > 0$ implies that $F_\tau(x; \varepsilon) > 0$ for all $x \in [0,1)$, so $H(0; \varepsilon) > 0$. Hence, $g(0; \varepsilon) > 0$. By continuity, there exists a value $x^* \in (0,1)$ with $g(x^*; \varepsilon) = 0$, yielding an interior steady state $(\pi_1^\infty, \pi_2^\infty) = (x^*, F_2(x^*; \varepsilon))$. Finally, for any $p \in [0,1)$, $\frac{\partial F_\tau(p;\varepsilon)}{\partial \varepsilon} > 0$, so both $F_\tau(\cdot; \varepsilon)$ shift up pointwise in ε, and thus so does $H(\cdot; \varepsilon)$. As $H(\cdot; \varepsilon)$ is linear-fractional, $H(x; \varepsilon) - x$ is quadratic with root $x = 1$, hence there is at most one additional (interior) fixed point. Because the interior fixed point is unique, it follows that $x^*(\varepsilon)$ is strictly increasing in ε, and therefore $\pi_2^\infty(\varepsilon) = F_2(x^*(\varepsilon); \varepsilon)$ is strictly increasing as well, proving statement (v).

As $\eta_\tau = 1$ for $\tau = 1, 2$ when $\mu = -1$, the final statement follows immediately from evaluating (9) at $\eta_\tau = 1$. $\square$

Proof of Proposition 4.

Proof. Fix $s(e_i, e_j) = 0$ for all e_i, e_j and $\beta(b_i, b_j) = 0$ for all b_i, b_j. Evaluating (15) at $\bar{\pi} = 1$ immediately gives $1 = 1$, so the biased steady state always exists.

To prove statements (i)–(iii), suppose that $\varepsilon = 0$ and $\mu > -1$. At $\varepsilon = 0$, $\gamma_1 = \gamma_2 = 1$, and evaluating at $\bar{\pi} = 0$ gives $0 = 0$, so the unbiased steady state exists, proving statement (i). As $\theta(0) = 0$ and both the unbiased and biased steady states exist, there are no additional steady states. Additionally,

$$\Delta \pi_\tau = (1 - \eta_\tau)(\theta(\bar{\pi}) - \pi_\tau). \tag{19}$$

Both types are pulled each period toward the same target $\theta(\bar{\pi})$. Hence, if $\pi_1 > \pi_2$, the higher type decreases relative to the lower whenever $\theta(\bar{\pi}) \leq \pi_1$, and the lower type increases relative to the higher whenever $\theta(\bar{\pi}) \geq \pi_2$. Thus, the dynamics pull π_1 and π_2 toward the diagonal $\pi_1 = \pi_2$; in particular, any steady state must satisfy $\pi_1 = \pi_2$ and thus $\bar{\pi} = \theta(\bar{\pi})$. For all $\bar{\pi} > 0$, the convexity of $\theta(\bar{\pi})$ coupled with $\theta(0) = 0$ implies that $\theta(\bar{\pi}) - \bar{\pi} < 0$, so there is convergence from almost all initial conditions to the unbiased steady state, , proving statement (ii).[23]

[23]If $\theta(\bar{\pi})$ is concave, then by an analogous argument, $\theta(\bar{\pi}) - \bar{\pi} > 0$, so there is convergence from almost all initial conditions to the biased steady state.

Now suppose $\theta(\bar{\pi}) = \bar{\pi}$. The law of motion (13) reduces to

$$\begin{pmatrix} \pi_1^{t+1} \\ \pi_2^{t+1} \end{pmatrix} = \underbrace{\begin{pmatrix} 1-(1-\eta_1)(1-q) & (1-\eta_1)(1-q) \\ (1-\eta_2)q & 1-(1-\eta_2)q \end{pmatrix}}_{\equiv \mathbf{Q}} \begin{pmatrix} \pi_1^t \\ \pi_2^t \end{pmatrix},$$

with eigenvalues 1 and $(1-q)\eta_1 + q\eta_2 \in (0,1)$ for $\mu > -1$. A direct calculation confirms that $q(1-\eta_2)\pi_1^{t+1} + (1-q)(1-\eta_1)\pi_2^{t+1} = q(1-\eta_2)\pi_1^t + (1-q)(1-\eta_1)\pi_2^t$, so the linear combination on the left is constant at its initial value, and since the second eigenvalue lies in $(0,1)$, the solution converges to a multiple of the unit eigenvector $(1,1)^T$ so $\pi_1^t - \pi_2^t \to 0$, proving statement (iii).

To prove statements (iv) and (v), suppose that $\varepsilon > 0$ and $\mu > -1$. In this case $(\gamma_1, \gamma_2) \in (0,1)^2$. Now suppose that $\theta(\bar{\pi})$ is linear. At $\pi_1^0 = \pi_2^0 = 0$ (so $\bar{\pi} = 0$),

$$[q\gamma_1 + (1-q)\gamma_2](0-1) + 1 > 0.$$

As the biased steady state exists, it immediately follows (by an analogous argument to statement (ii)) that all initial conditions converge to the biased steady state, proving statement (iv).[24]

Now suppose that $\theta(\bar{\pi})$ is convex and $[q\gamma_1 + (1-q)\gamma_2]\frac{d\theta(1)}{d\bar{\pi}} > 1$. Define $H(\bar{\pi}) \equiv [q\gamma_1 + (1-q)\gamma_2](\theta(\bar{\pi}) - 1) + 1$ and $g(\bar{\pi}) \equiv H(\bar{\pi}) - \bar{\pi}$. As $\varepsilon > 0$ and $(\gamma_1, \gamma_2) \in (0,1)^2$, $g(0) > 0$ and $g(1) = 0$. The assumed condition gives $\frac{dg(1)}{d\bar{\pi}} > 0$, so $g < 0$ in a left-neighborhood of 1; by the intermediate value theorem there exists $\bar{\pi}^* \in (0,1)$ with $g(\bar{\pi}^*) = 0$. The convexity of θ implies convexity of g, so $\bar{\pi}^*$ is unique and $\frac{dg(\bar{\pi}^*)}{d\bar{\pi}} < 0$, establishing global stability. Uniqueness of the constituent π_1^*, π_2^* follows because the right-hand side of (14) depends on π_τ only through $\bar{\pi}^*$, proving statement (v). If the slope condition fails, $g(\bar{\pi}) > 0$ for all $\bar{\pi} < 1$ and the biased steady state is the unique globally stable steady state.

If $\mu = -1$, then as is the case in Proposition 3, evaluating (13) at $\mu = -1$ (so $\eta_\tau = 1$ for $\tau = 1, 2$) yields the last statement. □

Proof of Proposition 5.

Proof. Fix $s(e_i, e_j) = 0$ for all e_i, e_j and $\beta(b_i, b_j) = 0$ for all b_i, b_j. By the exact same argument as Proposition 4, the biased steady state exists. Now, suppose for the moment that $\varepsilon = 0$ and $\mu > -1$. Again, by the same argument as Proposition 4, the unbiased steady state exists when $\varepsilon = 0$, as these arguments are made irrespective of the shape of $\theta(\bar{\pi})$. Define

$$H(\bar{\pi}) = [q\gamma_1 + (1-q)\gamma_2](\theta(\bar{\pi}) - 1) + 1$$

as the right-hand side of (15) and $g(\bar{\pi}) = H(\bar{\pi}) - \bar{\pi}$. Recall that $\theta(\cdot)$ is S-shaped with $\lim_{\bar{\pi}\to 0^+} \frac{dH(\bar{\pi})}{d\bar{\pi}} < 1$ and $\lim_{\bar{\pi}\to 1^-} \frac{dH(\bar{\pi})}{d\bar{\pi}} < 1$, and both biased and unbiased steady states exist. When $\varepsilon = 0$, $\gamma_1 = \gamma_2 = 1$, so $H(\bar{\pi}) = \theta(\bar{\pi})$. As $H(0) = 0$ and $\lim_{\bar{\pi}\to 0^+} \frac{dH(\bar{\pi})}{d\bar{\pi}} < 1$, $H(\bar{\pi}) < \bar{\pi}$ for $\bar{\pi} > 0$ sufficiently small. As $H(1) = 1$ and $\lim_{\bar{\pi}\to 1^-} \frac{dH(\bar{\pi})}{d\bar{\pi}} < 1$, $H(\bar{\pi}) > \bar{\pi}$ for $\bar{\pi} < 1$ sufficiently close to 1. By continuity, there exists an interior fixed point $\bar{\pi}_U^* \in (0,1)$ such that $\frac{dH(\bar{\pi}_U^*)}{d\bar{\pi}} > 1$ with the usual property that for all $\bar{\pi}^0 < \bar{\pi}_U^*$, there is convergence to the unbiased steady state and for all $\bar{\pi}^0 > \bar{\pi}_U^*$, there is convergence to the biased steady state. Uniqueness of $\bar{\pi}_U^*$ follows from the S-shape of θ, g has exactly one local minimum followed by one local maximum on $(0,1)$, and since $g(0) = 0$, $\frac{dg(0)}{d\bar{\pi}} < 0$, $g(1) = 0$, and $\frac{dg(1)}{d\bar{\pi}} < 0$, g can cross zero at most once on $(0,1)$.

[24] Allowing weak concavity does not affect this argument.

Uniqueness of the constituent π^*_{1U} and π^*_{2U} that form $\bar{\pi}^*_U$ follows from (14) and the argument in Proposition 4.

Now, suppose that $\varepsilon > 0$ and $\mu > -1$. Then, $H(0) = 1 - [q\gamma_1 + (1-q)\gamma_2] > 0$ (as $(\gamma_1, \gamma_2) \in (0,1)^2$), so the unbiased steady state no longer exists. As $\frac{dH(\bar{\pi})}{d\bar{\pi}} = [q\gamma_1 + (1-q)\gamma_2]\frac{d\theta(\bar{\pi})}{d\bar{\pi}}$, additional interior steady states exist iff $g(\bar{\pi}) < 0$ for some $\bar{\pi} \in (0,1)$. Because $g(0) = 1 - [q\gamma_1 + (1-q)\gamma_2] > 0$ and $g(1) = 0$, this condition requires that $\frac{dg(\bar{\pi})}{d\bar{\pi}}$ becomes positive somewhere, i.e., $\frac{d\theta(\bar{\pi})}{d\bar{\pi}} > [q\gamma_1 + (1-q)\gamma_2]^{-1}$ for some $\bar{\pi}$. Interior steady states therefore exist iff $\frac{d\theta(\bar{x})}{d\bar{\pi}} > [q\gamma_1 + (1-q)\gamma_2]^{-1}$, where $\bar{x}$ maximizes $\frac{d\theta(\bar{\pi})}{d\bar{\pi}}$ on $(0,1)$. As θ is single S-shaped, g has exactly one local minimum followed by one local maximum on $(0,1)$; together with $g(0) > 0$, $g(1) = 0$, and $\frac{dg(1)}{d\bar{\pi}} < 0$, this permits exactly two interior zeros when the condition holds. If this condition holds, there are three steady states: a stable interior steady state $\bar{\pi}^*$, an unstable interior tipping point $\bar{\pi}^*_U > \bar{\pi}^*$, and the biased steady state, with $g(\bar{\pi}) > 0$ on $(0, \bar{\pi}^*)$, $g(\bar{\pi}) < 0$ on $(\bar{\pi}^*, \bar{\pi}^*_U)$, and $g(\bar{\pi}) > 0$ on $(\bar{\pi}^*_U, 1)$, giving the respective basins of attraction. If this condition fails, $g(\bar{\pi}) > 0$ for all $\bar{\pi} \in [0,1)$ and the biased steady state is the unique steady state. The final statement for $\mu = -1$ follows as in Proposition 4. □

Proof of Proposition 6.

Proof. Suppose $s(e_i, e_j) = 0$ for all e_i, e_j, $\beta(b_i, b_j) = 0$ for all b_i, b_j, $\varepsilon > 0$, and $\theta(\bar{\pi})$ is S-shaped such that there are three steady states (as defined in the statement and proof of Proposition 5). Given $\mu \in \left(-1, \frac{1-q}{q}\right]$, let $\bar{\pi}^*_U(\mu)$ denote the unstable interior steady state such that $\frac{dH(\bar{\pi}^*_U(\mu))}{d\bar{\pi}} > 1$.

Note that

$$\gamma_\tau = \frac{1-\eta_\tau}{1-\eta_\tau(1-\varepsilon)}$$

is decreasing in η_τ and thus increasing in μ. Therefore, if $H(\bar{\pi}^*_U(\mu)) = \bar{\pi}^*_U(\mu)$, then for all $\mu' > \mu$, evaluating γ_1 and γ_2 at μ' and $\bar{\pi} = \bar{\pi}^*_U(\mu)$ yields

$$[q\gamma(\eta_1(\mu'), \varepsilon) + (1-q)\gamma(\eta_2(\mu'), \varepsilon)](\theta(\bar{\pi}^*_U(\mu)) - 1) + 1 < \bar{\pi}^*_U(\mu),$$

so $\bar{\pi}^*_U(\mu') > \bar{\pi}^*_U(\mu)$.

I now verify that $\bar{\pi}^*(\mu)$ is strictly decreasing in μ. At a stable interior steady state, $g(\bar{\pi}^*(\mu)) = 0$ (as defined in the proof of Proposition 5) and $\frac{\partial g(\bar{\pi})}{\partial \bar{\pi}} < 0$. Differentiating implicitly yields

$$\frac{d\bar{\pi}^*}{d\mu} = -\frac{\partial g/\partial \mu}{\partial g/\partial \bar{\pi}^*}.$$

Because $g(\bar{\pi}) = [q\gamma_1 + (1-q)\gamma_2](\theta(\bar{\pi}) - 1) + 1 - \bar{\pi}$ and $\theta(\bar{\pi}) - 1 < 0$ for $\bar{\pi} \in (0,1)$, an increase in μ lowers g pointwise, so $\frac{\partial g(\bar{\pi})}{\partial \mu} < 0$. Combined with $\frac{\partial g(\bar{\pi}^*)}{\partial \bar{\pi}} < 0$ at a stable fixed point, $\frac{d\bar{\pi}^*}{d\mu} < 0$. Hence, $\bar{\pi}^*(\mu)$ is strictly decreasing in μ.

Now, consider an initial $\mu = \mu^0$ and $\bar{\pi}^0 > \bar{\pi}^*_U(\mu^0)$. In this case, under $\mu = \mu^0$, the trajectory remains in the basin of attraction of the biased steady state and converges toward it. As $\bar{\pi}^*_U(\mu)$ is continuous in μ (by the implicit function theorem, since $\frac{dH(\bar{\pi}^*_U(\mu))}{d\bar{\pi}} > 1$ implies $\frac{\partial g(\bar{\pi}^*_U)}{\partial \bar{\pi}} \neq 0$) and strictly increasing in μ, and $\bar{\pi}^0 < \bar{\pi}^*_U(\frac{1-q}{q})$ by hypothesis, there exists $\mu' \in (\mu^0, \frac{1-q}{q})$ such that $\bar{\pi}^0 < \bar{\pi}^*_U(\mu')$, so at $\mu = \mu'$, $\bar{\pi}^0$ is in the basin of attraction for the lower steady state.

Under $\mu = \mu'$, the trajectory converges to the lower interior steady state, so there exists $t' > 0$ such that $\bar{\pi}^{t'} < \bar{\pi}^*_U(\mu^0)$ (as $\bar{\pi}^*(\mu') < \bar{\pi}^*(\mu^0) < \bar{\pi}^*_U(\mu^0)$, where the first inequality follows from $\frac{d\bar{\pi}^*}{d\mu} < 0$ and $\mu' > \mu^0$, and the second from Proposition 5). Therefore, returning μ to μ^0 implies that there will be persistent convergence of $\bar{\pi}^t$ to $\bar{\pi}^*(\mu^0)$ (the smallest interior steady state). □

Proof of Lemma 1.

Proof. Recall from Section 2.2 that

$$\rho_{b_i,b_j} = s^*(b_i, b_j) + (1 - s^*(b_i, b_j))[\beta(b_i, b_j) + (1 - \beta(b_i, b_j))(1 - \theta(\bar{\pi}))].$$

Because efforts are symmetric in the equilibrium of the social interaction stage game, letting $e_i^*(b_i, b_j) = e_j^*(b_i, b_j) = e^*$ (suppressing the bias values),

$$s^*(b_i, b_j) = e^* + e^* - e^* \times e^* = 1 - (1 - e^*)^2.$$

Now, consider the probability of a negative interaction between i and j:

$$\begin{aligned}1 - \rho_{b_i,b_j} &= 1 - s^*(b_i, b_j) - (1 - s^*(b_i, b_j))[\beta(b_i, b_j) + (1 - \beta(b_i, b_j))(1 - \theta(\bar{\pi}))\\ &= (1 - e^*)^2(1 - \beta(b_i, b_j))\theta(\bar{\pi}),\end{aligned}$$

so

$$\rho_{b_i,b_j} = 1 - (1 - e^*)^2(1 - \beta(b_i, b_j))\theta(\bar{\pi}),$$

Substituting (5) for e^* yields

$$\rho_{b_i,b_j} = 1 - \frac{(1 - \beta(b_i, b_j))\theta(\bar{\pi})}{[1 + (1 - \beta(b_i, b_j))\theta(\bar{\pi})V]^2}.$$

Differentiating this expression with respect to $\bar{\pi}$ yields

$$\frac{\partial \rho_{b_i,b_j}}{\partial \bar{\pi}} = -\frac{d\theta(\bar{\pi})}{d\bar{\pi}} \times \frac{(1 - \beta(b_i, b_j))[1 - (1 - \beta(b_i, b_j))\theta(\bar{\pi})V]}{[1 + (1 - \beta(b_i, b_j))\theta(\bar{\pi})V]^3}.$$

As $\frac{d\theta(\bar{\pi})}{d\bar{\pi}} > 0$ and $(1 - \beta(b_i, b_j) \in 0, 1]$, $\frac{\partial \rho_{b_i,b_j}}{\partial \bar{\pi}}$ is monotonic in $\bar{\pi}$ if and only if

$$1 - (1 - \beta(b_i, b_j))\theta(\bar{\pi})V$$

is monotonic; i.e., if at $\bar{\pi} = 1$,

$$\begin{aligned}&1 - (1 - \beta(b_i, b_j))V \geq 0\\ &\iff V \leq \frac{1}{1 - \beta(b_i, b_j)} \equiv \hat{V},\end{aligned}$$

proving the Lemma. □

Proof of Lemma 2.

Proof. Suppose that $\mu > -1$. Recall by (6) that

$$\Delta\pi_\tau = (1 - \eta_\tau)\left[1 - \bar{\rho}_\tau(\pi) - \pi_\tau\right] + \eta_\tau(1 - \pi_\tau)\varepsilon,$$

which when evaluated at the biased steady state $(\pi_1, \pi_2) = (1, 1)$ yields

$$\Delta\pi_\tau = -(1 - \eta_\tau)\bar{\rho}_\tau(1).$$

Hence, for $\mu > -1$ (so $\eta_\tau < 1$), the biased steady state exists if and only if $\bar{\rho}_\tau(1) = 0$, where by the proof of Lemma 1,

$$\bar{\rho}_\tau(1) = \rho_{1,1} = 1 - \frac{\theta(1)}{[1 + \theta(1)V]^2} = 1 - \frac{1}{(1 + V)^2} = \frac{V(2 + V)}{(1 + V)^2} > 0.$$

Therefore, the biased steady state does not exist.

Now, evaluating (6) at the unbiased steady state $(\pi_1, \pi_2) = (0, 0)$ yields

$$\Delta\pi_\tau = -(1 - \eta_\tau)(1 - \bar{\rho}_\tau(0)) - \eta_\tau\varepsilon.$$

Hence, for $\mu > -1$, the unbiased steady state exists if and only if $\bar{\rho}_\tau(0) = 1$ and $\varepsilon = 0$, where by the proof of Lemma 1,

$$\bar{\rho}_\tau(0) = \rho_{0,0} = 1 - \frac{(1 - \beta_0)\theta(0)}{[1 + (1 - \beta_0)\theta(0)V]^2} = 1$$

Thus, the unbiased steady state exists whenever $\varepsilon = 0$ and $\mu > -1$. □

Proof of Proposition 7.

Proof. Fix $\mu > -1$. Recall by (6) that

$$\Delta\pi_\tau = (1 - \eta_\tau)[1 - \bar{\rho}_\tau(\pi) - \pi_\tau] + \eta_\tau(1 - \pi_\tau)\varepsilon$$

where $\bar{\rho}_\tau(\pi)$ is the (type-τ) expected probability of a positive interaction under equilibrium effort.

Fix a realized pair of biases (b_i, b_j) and write $\beta \equiv \beta(b_i, b_j) \in [0, 1]$ and $\theta \equiv \theta(\bar{\pi})$. From Proposition 1, equilibrium effort satisfies

$$1 - s^*(b_i, b_j) = (1 - e^*(b_i, b_j))^2 = \frac{1}{(1 + (1 - \beta)\theta V)^2},$$

so the match-level negative-interaction probability is

$$m_\beta(\bar{\pi}) \equiv \Pr(\text{negative} \mid b_i, b_j) = \frac{(1 - \beta)\theta(\bar{\pi})}{(1 + (1 - \beta)\theta(\bar{\pi})V)^2}. \tag{20}$$

Equivalently, $\rho_{b_i,b_j}(\bar{\pi}) = 1 - m_\beta(\bar{\pi})$.

Define

$$\tilde{p}'(\bar{\pi}) := \frac{d\theta(\bar{\pi})}{d\bar{\pi}} \times \max_{\beta\in[0,1]} \frac{(1 - \beta)|1 - (1 - \beta)\theta(\bar{\pi})V|}{(1 + (1 - \beta)\theta(\bar{\pi})V)^3}.$$

Define $z = (1 - \beta)\theta(\bar{\pi})V$, so the above can be represented by the equivalent optimization problem

$$\frac{1}{\theta(\bar{\pi})V} \times \max_{z\in[0,\theta(\bar{\pi})V]} \frac{z|1 - z|}{(1 + z)^3}.$$

Solving this maximization problem yields

$$(1-\beta)\theta V = z^* = \begin{cases} 2-\sqrt{3} & \text{if } \theta(\bar{\pi})V \leq 2+\sqrt{3} \\ \\ 2+\sqrt{3} & \text{if } \theta(\bar{\pi})V > 2+\sqrt{3}. \end{cases}$$

As $\frac{z^*|1-z^*|}{(1+z^*)^3} = \frac{\sqrt{3}}{18}$ for both $z \leq 1$ and $z > 1$, $\left|\frac{dm_\beta}{d\bar{\pi}}\right| \leq \tilde{p}'(\bar{\pi})$. Define $\tilde{P}' := \sup_{\bar{\pi}\in[0,1]} \tilde{p}'(\bar{\pi})$.

Now, denote by

$$\bar{\Delta} = \sup_{\bar{\pi}\in[0,1]} \max_{\beta,\beta'\in\{0,\beta_1,\beta_0\}} |m_\beta(\bar{\pi}) - m_{\beta'}(\bar{\pi})| \geq 0$$

the largest gap between the $m_\beta(\bar{\pi})$ terms across all bias configurations. Expanding $M_\tau(\pi) = 1 - \bar{\rho}_\tau(\pi)$ yields

$$M_\tau(\pi) = (1-\pi_\tau)(1-\pi_{-\tau})m_{\beta_0} + [(1-\pi_\tau)\pi_{-\tau} + \pi_\tau(1-\pi_{-\tau})]m_{\beta_1} + \pi_\tau\pi_{-\tau}m_{\beta_2}.$$

Differentiating this expression and letting $q_\tau = \lambda(I_\tau)$ yields

$$\frac{\partial M_\tau}{\partial \pi_\tau} = \underbrace{(1-\pi_{-\tau})(m_{\beta_1} - m_{\beta_0}) + \pi_{-\tau}(m_{\beta_2} - m_{\beta_1})}_{W_\tau^{\text{own}}} + q_\tau \frac{\partial M_\tau}{\partial \bar{\pi}}$$

$$\frac{\partial M_\tau}{\partial \pi_{-\tau}} = \underbrace{(1-\pi_\tau)(m_{\beta_1} - m_{\beta_0}) + \pi_\tau(m_{\beta_2} - m_{\beta_1})}_{W_\tau^{\text{cross}}} + q_{-\tau} \frac{\partial M_\tau}{\partial \bar{\pi}}.$$

Each W_τ^{own} is a convex combination of $(m_{\beta_1} - m_{\beta_0})$ and $(m_{\beta_2} - m_{\beta_1})$, so $|W_\tau^{\text{own}}| \leq \bar{\Delta}$. Likewise, $|W_\tau^{\text{cross}}| \leq \bar{\Delta}$. By the triangle inequality and the fact that $\left|\frac{dm_\beta}{d\bar{\pi}}\right| \leq \tilde{p}'(\bar{\pi})$,

$$\sum_{k=1}^{2} \left|\frac{\partial M_\tau}{\partial \pi_k}\right| \leq |W_\tau^{own}| + |W_\tau^{cross}| + \sum_{k=1}^{2} q_k \left|\frac{\partial M_\tau}{\partial \bar{\pi}}\right| \leq 2\bar{\Delta} + \tilde{P}'. \tag{21}$$

At a steady state, (6) implies

$$0 = (1-\eta_\tau)[M_\tau(\pi) - \pi_\tau] + \eta_\tau(1-\pi_\tau)\varepsilon,$$

which can be rearranged to

$$\pi_\tau = \frac{(1-\eta_\tau)M_\tau(\pi) + \eta_\tau\varepsilon}{1-\eta_\tau(1-\varepsilon)} \equiv T_\tau(\pi), \tag{22}$$

$\tau \in \{1,2\}$. Thus, steady states are fixed points of the mapping $T = (T_1, T_2) : [0,1]^2 \to [0,1]^2$. Because $T_\tau(\pi) = \gamma_\tau M_\tau(\pi) + (1-\gamma_\tau)$,

$$\sum_{k=1}^{2} \left|\frac{\partial T_\tau}{\partial \pi_k}\right| = \gamma(\eta_\tau, \varepsilon) \sum_{k=1}^{2} \left|\frac{\partial M_\tau}{\partial \pi_k}\right| \leq \gamma(\eta_\tau, \varepsilon)\big(2\bar{\Delta} + \tilde{P}'\big). \tag{23}$$

As $\gamma(\eta_\tau, \varepsilon)$ is strictly decreasing in η and $\eta_1 \geq \eta_2$, the maximum row sum is attained at $\tau = 2$. Now, assume that $2\bar{\Delta} + \tilde{P}' < \gamma(\eta_2, \varepsilon)^{-1}$. Then (23) implies that

$$\max_{\tau\in\{1,2\}} \sum_{k=1}^{2} \left|\frac{\partial T_\tau}{\partial \pi_k}\right| < 1$$

uniformly on $[0,1]^2$, so T is a global contraction. By Banach's fixed-point theorem, T has a unique fixed point in $[0,1]^2$. Lemma 2 rules out $(1,1)$ for $\mu > -1$, so the unique steady state lies in $[0,1)^2$, proving statement (i).

Lastly, assume $\varepsilon > 0$ and that $\frac{dm_{\beta_0}(\bar{\pi})}{d\bar{\pi}}$ is single-peaked on $(0,1)$, attaining its unique maximum at $\bar{\pi}_M \in (0,1)$. Let $M_\tau(\pi) = m_{\beta_0} + R_\tau(\pi)$, where $|R_\tau(\pi)| \le \bar{\Delta}$ for all $\pi \in [0,1]^2$ and $\bar{\gamma} = q\gamma_1 + (1-q)\gamma_2$. Define the aggregate map

$$\Psi(\bar{\pi}) = \bar{\gamma} m_{\beta_0}(\bar{\pi}) + \sum_\tau q_\tau \frac{\eta_\tau \varepsilon}{1 - \eta_\tau(1-\varepsilon)} > 0,$$

so that at any steady state, $\bar{\pi} = \Psi(\bar{\pi}) + \mathcal{E}(\pi)$ where $\mathcal{E}(\pi)$ is an error term such that $|\mathcal{E}(\pi)| \le \bar{\Delta}$ uniformly. As $\frac{d\Psi}{d\bar{\pi}} = \bar{\gamma} \times \frac{dm_{\beta_0}}{d\bar{\pi}}$ and $\frac{dm_{\beta_0}}{d\bar{\pi}}$ is single-peaked at $\bar{\pi}_M$ by hypothesis, $\frac{d\Psi}{d\bar{\pi}}$ is also single-peaked at $\bar{\pi}_M$. The conditions

$$\frac{dm_{\beta_0}(\bar{\pi}_M)}{d\bar{\pi}} > \bar{\gamma}^{-1}, \qquad \tilde{p}'(0), \tilde{p}'(1) < \bar{\gamma}^{-1}$$

imply that $\frac{d\Psi}{d\bar{\pi}} < 1$ near $\bar{\pi} = 0$ and $\bar{\pi} = 1$ (using $\frac{dm_{\beta_0}}{d\bar{\pi}} \le \tilde{p}'$ at the endpoints), while $\frac{d\Psi}{d\bar{\pi}} > 1$ at $\bar{\pi}_M$. As

$$\Psi(0) = \sum_\tau q_\tau \frac{\eta_\tau \varepsilon}{1 - \eta_\tau(1-\varepsilon)}$$

Ψ lies above the diagonal near 0 and $\Psi(1) < 1$ (as $m_{\beta_0}(1) < 1$ and $\bar{\gamma} < 1$), the graph of Ψ crosses the 45-degree line exactly three times at $\bar{\pi}_1^* < \bar{\pi}_2^* < \bar{\pi}_3^*$.

For each $\bar{\pi}_k^*$, the fixed-point equations $\pi_\tau = T_\tau(\pi)$ subject to $q\pi_1 + (1-q)\pi_2 = \bar{\pi}_k^*$ reduce to a single equation in one free variable. By the monotonicity and continuity of T_τ, this equation has at least one solution, yielding a steady state $\pi^{(k)} \in (0,1)^2$.

More precisely, at each crossing $\bar{\pi}_k^*$, define the two-dimensional map $\Phi(\pi) = T(\pi) - \pi$. At $\bar{\Delta} = 0$ (i.e., $\beta_1 = \beta_0$ and $\beta_2 = \beta_0$), the system reduces exactly to the one-dimensional map. $M_\tau(\pi)$ depends on π only through $\bar{\pi}$, so the fixed-point equations collapse to $\bar{\pi} = \Psi(\bar{\pi})$ together with $\pi_\tau = \gamma_\tau m_{\beta_0}(\bar{\pi}) + (1-\gamma_\tau)$. Each of the three aggregate crossings then yields a unique $(\pi_1^{(k)}, \pi_2^{(k)})$ at which $\Phi = 0$. At the two stable crossings $(k = 1, 3)$, $\frac{d\Psi}{d\bar{\pi}} < 1$, and at the unstable crossing $(k = 2)$, $\frac{d\Psi}{d\bar{\pi}} > 1$. In each case $D\Phi$ is nonsingular at the corresponding fixed point. By the implicit function theorem, each fixed point persists under small perturbations of the bias parameters. As $\bar{\Delta}$ is continuous in $(\beta_0, \beta_1, \beta_2)$ and equals zero when $\beta_0 = \beta_1 = \beta_2$, the three fixed points persist for all $\bar{\Delta} < \hat{\Delta}$ for some $\hat{\Delta} > 0$.[25]

Let $g(\bar{\pi}) = \Psi(\bar{\pi}) - \bar{\pi}$. By the above analysis, g has exactly three zeros at $\bar{\pi}_1^* < \bar{\pi}_2^* < \bar{\pi}_3^*$. On the interval $[\bar{\pi}_1^*, \bar{\pi}_2^*]$, g is strictly negative (as Ψ crosses the diagonal from above at $\bar{\pi}_1^*$ and from below at $\bar{\pi}_2^*$). The error $\mathcal{E}(\pi)$ perturbs each crossing level by at most $\bar{\Delta}$. Define

$$d_1 = - \min_{\bar{\pi} \in [\bar{\pi}_1^*, \bar{\pi}_2^*]} g(\bar{\pi}) > 0.$$

[25] The binding constraint is not proximity of β_1 to β_0 per se, but rather that the amplitude of the S-shaped nonlinearity in Ψ exceeds the perturbation introduced by heterogeneous β values. Concretely, $\hat{\Delta}$ is bounded below by $\min_k \left|1 - \frac{d\Psi(\bar{\pi}_k^*)}{d\bar{\pi}}\right|$ scaled by a Lipschitz constant. When the three crossings of Ψ with the diagonal are well separated and steep, the fixed points survive large differences between β_0 and β_1. The condition $\bar{\Delta} < \hat{\Delta}$ is therefore least restrictive precisely when tipping dynamics are most pronounced.

Similarly, on $[\bar{\pi}_2^*, \bar{\pi}_3^*]$, g is strictly positive. Define

$$d_2 = \max_{\bar{\pi} \in [\bar{\pi}_2^*, \bar{\pi}_3^*]} g(\bar{\pi}) > 0.$$

The full steady-state system requires $g(\bar{\pi}) = \mathcal{E}(\pi)$, where $|\mathcal{E}(\pi)| \leq \bar{\gamma}\bar{\Delta}$ uniformly.

Assume $\bar{\Delta} < \min(d_1/\bar{\gamma}, d_2/\bar{\gamma}, (1-\bar{\gamma})/\bar{\gamma}, 1 - m_{\beta_0}(1)) \equiv \hat{\Delta}$. At $\bar{\pi} = 0$, $g(0) = 1 - \bar{\gamma} > 0$. At the minimizer $\bar{\pi}_{\min} \in (\bar{\pi}_1^*, \bar{\pi}_2^*)$, $g(\bar{\pi}_{\min}) = -d_1 < -\bar{\gamma}\bar{\Delta}$. Hence, for any $\mathcal{E}$ with $|\mathcal{E}| \leq \bar{\gamma}\bar{\Delta}$, the $g(\bar{\pi}) - \mathcal{E}$ satisfies $g(0) - \mathcal{E} \geq (1-\bar{\gamma}) - \bar{\gamma}\bar{\Delta} > 0$ and $g(\bar{\pi}_{\min}) - \mathcal{E} \leq -d_1 + \bar{\gamma}\bar{\Delta} < 0$. By the intermediate value theorem, there exists a root $\hat{\pi}_1 \in (0, \bar{\pi}_{\min})$. At the maximizer $\bar{\pi}_{\max} \in (\bar{\pi}_2^*, \bar{\pi}_3^*)$, $g(\bar{\pi}_{\max}) = d_2 > \bar{\gamma}\bar{\Delta}$. As $g(\bar{\pi}_{\min}) - \mathcal{E} < 0$ and $g(\bar{\pi}_{\max}) - \mathcal{E} > d_2 - \bar{\gamma}\bar{\Delta} > 0$, $g(\bar{\pi}) - \mathcal{E}$ changes sign on $[\bar{\pi}_{\min}, \bar{\pi}_{\max}]$, yielding a root $\hat{\pi}_2$. Finally, $g(1) = \Psi(1) - 1 < 0$ (as $m_{\beta_0}(1) < 1$ and $\bar{\gamma} < 1$), so $g(1) - \mathcal{E} \leq g(1) + \bar{\gamma}\bar{\Delta}$. Because $g(1) = -\bar{\gamma}(1 - m_{\beta_0}(1)) < 0$, this remains strictly negative for $\bar{\Delta} < 1 - m_{\beta_0}(1)$. As $g(\bar{\pi}_{\max}) - \mathcal{E} > 0$, $g(\bar{\pi}) - \mathcal{E}$ changes sign on $[\bar{\pi}_{\max}, 1]$, yielding a third root $\hat{\pi}_3$.

For each aggregate crossing $\hat{\pi}_k$, the fixed-point equations $\pi_\tau = T_\tau(\pi)$ subject to $q\pi_1 + (1-q)\pi_2 = \hat{\pi}_k$ reduce to a single equation in one free variable. By the monotonicity and continuity of T_τ, and because T_τ maps $[0,1]^2$ into $(0,1)^2$ (since $\varepsilon > 0$ and $V > 0$), this equation has at least one solution, yielding a steady state $\pi^{(k)} \in (0,1)^2$. The three crossings $\hat{\pi}_1 < \hat{\pi}_2 < \hat{\pi}_3$ are distinct, so the corresponding steady states are also distinct, proving statement (ii). □

Proof of Proposition 8.

Proof. From the proof of Proposition 7, the aggregate map is

$$\Psi(\bar{\pi}; \mu) = \bar{\gamma}(\mu) m_{\beta_0}(\bar{\pi}) + \sum_\tau q_\tau \frac{\eta_\tau(\mu)\varepsilon}{1 - \eta_\tau(\mu)(1-\varepsilon)}.$$

As $1 - \gamma_\tau = \frac{\eta_\tau \varepsilon}{(1 - \eta_\tau(1-\varepsilon))}$, summing gives $\sum_\tau q_\tau(1 - \gamma_\tau) = 1 - \bar{\gamma}$, so

$$\Psi(\bar{\pi}; \mu) = 1 - \bar{\gamma}(\mu)\big(1 - m_{\beta_0}(\bar{\pi})\big).$$

This is structurally identical to $H(\bar{\pi}) = 1 - \bar{\gamma}(1 - \theta(\bar{\pi}))$ in Proposition 6, replacing θ with m_{β_0}.

Define $g(\bar{\pi}; \mu) \equiv \Psi(\bar{\pi}; \mu) - \bar{\pi}$. Because $\bar{\gamma}(\mu)$ is strictly increasing in μ (since η_τ is strictly decreasing in μ and $\gamma_\tau = \frac{(1-\eta_\tau)}{(1-\eta_\tau(1-\varepsilon))}$ is strictly decreasing in η_τ), and $m_{\beta_0}(\bar{\pi}) < 1$ for all $\bar{\pi} \in [0,1]$ (as $V > 0$), it follows that

$$\frac{\partial g(\bar{\pi}; \mu)}{\partial \mu} = -\frac{d\bar{\gamma}}{d\mu}\big(1 - m_{\beta_0}(\bar{\pi})\big) < 0$$

for all $\bar{\pi} \in [0, 1)$.

At the unstable crossing $\bar{\pi}_2^*$, the map Ψ crosses the diagonal from below, so $\frac{\partial g(\bar{\pi};\mu)}{\partial \bar{\pi}} > 0$. The implicit function theorem applied to $g(\bar{\pi}_2^*(\mu); \mu) = 0$ yields

$$\frac{d\bar{\pi}_2^*}{d\mu} = -\frac{\partial g/\partial \mu}{\partial g/\partial \bar{\pi}} > 0,$$

so $\bar{\pi}_2^*(\mu)$ is strictly increasing in μ. At $\bar{\pi}_1^*$, the map Ψ crosses from above, so $\partial g/\partial \bar{\pi} < 0$. By the same argument, $\frac{d\bar{\pi}_1^*}{d\mu} < 0$, so $\bar{\pi}_1^*(\mu)$ is strictly decreasing in μ.

The remainder of the argument follows the proof of Proposition 6. As $\bar{\pi}_2^*(\mu)$ is continuous and strictly increasing in μ with $\bar{\pi}^0 < \bar{\pi}_2^*\left(\frac{1-q}{q}\right)$ by hypothesis, there exists $\mu' \in \left(\mu^0, \frac{1-q}{q}\right)$ such that $\bar{\pi}^0 < \bar{\pi}_2^*(\mu')$. Thus, $\bar{\pi}_1^*(\mu') < \bar{\pi}_1^*(\mu^0) < \bar{\pi}_2^*(\mu^0) < \bar{\pi}^0 < \bar{\pi}_2^*(\mu')$, and the trajectory from $\bar{\pi}^0$ under μ' lies in the basin of attraction of $\bar{\pi}_1^*(\mu')$ and converges toward it. Because $\bar{\pi}_1^*(\mu') < \bar{\pi}_1^*(\mu^0) < \bar{\pi}_2^*(\mu^0)$ (using strict decrease of $\bar{\pi}_1^*$ and the three-steady-state ordering), there exists $t' > 0$ such that $\bar{\pi}^{t'} < \bar{\pi}_2^*(\mu^0)$. Returning μ to μ^0 places the trajectory in the basin of attraction of $\bar{\pi}_1^*(\mu^0)$, yielding persistent convergence to $\bar{\pi}_1^*(\mu^0) < \bar{\pi}^0$. □

The following Lemma is useful for proving Proposition 9.

Lemma 6. *For all $\bar{\pi} \in [0,1]$ and $V \geq 0$, $\frac{dm_{\beta_0}(\bar{\pi})}{d\bar{\pi}} \leq \frac{d\theta(\bar{\pi})}{d\bar{\pi}}$, with strict inequality for all $\bar{\pi} > 0$ whenever $V > 0$ and $\beta_0 \in [0,1)$.*

Proof. Differentiating m_{β_0} yields

$$\frac{dm_{\beta_0}}{d\bar{\pi}} = \frac{d\theta(\bar{\pi})}{d\bar{\pi}} \times \frac{(1-\beta_0)[1-(1-\beta_0)\theta(\bar{\pi})V]}{[1+(1-\beta_0)\theta(\bar{\pi})V]^3}.$$

Set $z = (1-\beta_0)\theta(\bar{\pi})V \geq 0$ and define

$$A(z) = \frac{(1-\beta_0)(1-z)}{(1+z)^3},$$

so $\frac{dm_{\beta_0}}{d\bar{\pi}} = \frac{d\theta(\bar{\pi})}{d\bar{\pi}} \times A(z)$. I now show that $A(z) < 1$ for all $z > 0$. If $z \geq 1$, then $1 - z \leq 0$, so $A(z) \leq 0 < 1$. If $0 < z < 1$, then $(1-\beta_0) < 1$ and $(1-z) < 1 \leq (1+z)^3$, so $A(z) < 1$. At $z = 0$, $A(0) = 1 - \beta_0 \leq 1$, with equality only when $\beta_0 = 0$. Hence, $A(z) < 1$ for all $z > 0$, i.e., whenever $\theta(\bar{\pi}) > 0$, $V > 0$, and $\beta_0 \in [0,1)$. □

Proof of Proposition 9.

Proof. From the proofs of Propositions 6 and 8, both the societal-bias benchmark and the full model reduce a mapping relying only on $\bar{\pi}$:

$$H(\bar{\pi}) = 1 - \bar{\gamma}(1-\theta(\bar{\pi}))$$
$$\Psi(\bar{\pi}) = 1 - \bar{\gamma}(1-m_{\beta_0}(\bar{\pi})).$$

Multiple steady states in each model correspond to multiple crossings of the respective map above with the 45-degree line.

Multiple steady states in the societal-bias benchmark require

$$\sup_{\bar{\pi}\in[0,1]} \frac{d\theta(\bar{\pi})}{d\bar{\pi}} > \bar{\gamma}^{-1}, \tag{24}$$

while multiple steady states in the full model require

$$\sup_{\bar{\pi}\in[0,1]} \frac{dm_{\beta_0}(\bar{\pi})}{d\bar{\pi}} > \bar{\gamma}^{-1}. \tag{25}$$

By Lemma 6, for $V > 0$ and $\beta_0 \in [0,1)$, $\sup_{\bar{\pi}} \frac{dm_{\beta_0}}{d\bar{\pi}} < \sup_{\bar{\pi}} \frac{d\theta(\bar{\pi})}{d\bar{\pi}}$. To verify this claim, note that since θ is S-shaped and continuously differentiable, its maximal slope is attained at some interior $\bar{\pi}^* > 0$. Lemma 6 gives $\frac{dm_{\beta_0}(\bar{\pi}^*)}{d\bar{\pi}} < \frac{d\theta(\bar{\pi}^*)}{d\bar{\pi}}$, and because $A(z) < 1$ holds uniformly over all $\bar{\pi} > 0$, it follows that $\sup_{\bar{\pi}} \frac{dm_{\beta_0}}{d\bar{\pi}} < \sup_{\bar{\pi}} \frac{d\theta(\bar{\pi})}{d\bar{\pi}}$. Hence, the full-model multiplicity region is a subset of the benchmark region. Choosing $\bar{\gamma}^{-1} \in \big(\sup_{\bar{\pi}} \frac{dm_{\beta_0}}{d\bar{\pi}}, \sup_{\bar{\pi}} \frac{d\theta(\bar{\pi})}{d\bar{\pi}}\big)$, which is a nonempty open interval for any $V > 0$ and $\beta_0 \in [0, 1)$, yields parameters satisfying (24) but not (25), proving that this subset is strict.

The individual-bias channel provides a further constraint. The uniqueness bound of Proposition 7(i) requires $2\bar{\Delta} + \tilde{P}' < \gamma(\eta_2, \varepsilon)^{-1}$, where $\bar{\Delta} > 0$ whenever $\beta_1 < \beta_0$. This additional gap shrinks the parameter region supporting multiplicity beyond what Lemma 6 alone implies, proving statement (i).

A necessary condition for basin-switching in either model is the existence of multiple steady states. By statement (i), the parameter region supporting multiple steady states in the full model is strictly contained in that of the benchmark. Therefore, the parameter region supporting basin-switching is weakly, and strictly for $V > 0$ and $\beta_0 \in [0, 1)$, smaller in the full model. At parameter configurations where multiplicity holds in the benchmark but fails in the full model, the full model has a unique steady state and no basin-switching is possible.

It remains to compare the width of the basin-switching window $\big(\bar{\pi}_2^*(\mu^0), \bar{\pi}_2^*\big(\frac{1-q}{q}\big)\big)$ conditional on multiple steady states existing in both models. As $\Psi(\bar{\pi}) \leq H(\bar{\pi})$ for all $\bar{\pi}$ (with strict inequality for $\bar{\pi} > 0$), the aggregate map in the full model lies uniformly below that of the benchmark. For a given $\bar{\gamma}$, an unstable interior crossing of Ψ with the 45-degree line therefore occurs at a level $\bar{\pi}_2^{\Psi}$ that is weakly above the corresponding crossing $\bar{\pi}_2^{H}$ of H. Because $\Psi < H$, the map must travel further before re-crossing the diagonal from below, placing the unstable tipping point higher. The basin of attraction of the low-bias steady state is therefore no larger in the full model, and the window of initial conditions from which basin-switching is feasible is weakly narrower.

Therefore, the full-model basin-switching region, the intersection of the smaller multiplicity parameter set with the narrower window of initial conditions, is weakly, and for $V > 0$ and $\beta_0 \in [0, 1)$ strictly, smaller than in the societal-bias benchmark, proving statement (ii). □

Proof of Proposition 10.

Proof. From the proof of Proposition 8,

$$\Psi(\bar{\pi}; V) = 1 - \bar{\gamma}\big(1 - m_{\beta_0}(\bar{\pi}; V)\big).$$

Steady states of the full system correspond to fixed points of $\Psi(\cdot; V)$ up to the perturbation $\mathcal{E}(\pi)$ bounded by $\bar{\Delta}$, as established in the proof of Proposition 7.

Differentiating $m_{\beta_0}(\bar{\pi}; V)$ with respect to V yields

$$\frac{\partial m_{\beta_0}}{\partial V} = -\frac{2(1-\beta_0)^2\theta(\bar{\pi})^2}{[1 + (1-\beta_0)\theta(\bar{\pi})V]^3} < 0$$

for all $\bar{\pi} > 0$. Therefore

$$\frac{\partial \Psi(\bar{\pi}; V)}{\partial V} = \bar{\gamma}\frac{\partial m_{\beta_0}}{\partial V} < 0$$

for all $\bar{\pi} \in (0, 1]$. Define $g(\bar{\pi}; V) \equiv \Psi(\bar{\pi}; V) - \bar{\pi}$. At any stable steady state $\bar{\pi}_k^*(V)$, Ψ crosses the diagonal from above, so $\frac{\partial g(\bar{\pi})}{\partial \bar{\pi}} < 0$. As $\frac{\partial g(\bar{\pi})}{\partial V} < 0$ and $\frac{\partial g(\bar{\pi})}{\partial \bar{\pi}} < 0$ at $\bar{\pi}_1^*$, the implicit function theorem gives

$$\frac{d\bar{\pi}_1^*}{dV} = -\frac{\partial g/\partial V}{\partial g/\partial \bar{\pi}} < 0,$$

so $\bar{\pi}_1^*(V)$ is strictly decreasing in V.

From Lemma 6,

$$\frac{dm_{\beta_0}(\bar{\pi};V)}{d\bar{\pi}} = \frac{d\theta(\bar{\pi})}{d\bar{\pi}} \times A(z), \quad z = (1-\beta_0)\theta(\bar{\pi})V,$$

where $A(z) \to 0$ as $z \to \infty$. Fix $\delta > 0$. For all $\bar{\pi} \in [\delta, 1]$, $\theta(\bar{\pi}) \geq \theta(\delta) > 0$, so $z \geq (1-\beta_0)\theta(\delta)V \to \infty$, and $A(z) \to 0$ uniformly on $[\delta, 1]$ as $V \to \infty$. On $[0, \delta]$, by the hypothesis $\tilde{p}'(0) < \bar{\gamma}^{-1}$ in Proposition 7(ii), we have $\bar{\gamma}\frac{dm_{\beta_0}}{d\bar{\pi}} \leq \bar{\gamma}\tilde{p}'(\bar{\pi}) < 1$ near $\bar{\pi} = 0$ for any $V \geq V^0$. As $\delta \to 0$ this region vanishes. Hence,

$$\lim_{V\to\infty} \sup_{\bar{\pi}\in[0,1]} \frac{d\Psi(\bar{\pi};V)}{d\bar{\pi}} = \lim_{V\to\infty} \bar{\gamma} \sup_{\bar{\pi}\in[0,1]} \frac{dm_{\beta_0}(\bar{\pi};V)}{d\bar{\pi}} = 0.$$

The quantities $\tilde{P}'(V) = \sup_{\bar{\pi}} \tilde{p}'(\bar{\pi};V)$ and $\bar{\Delta}(V) = \sup_{\bar{\pi}}[p(\bar{\pi};V) - m_{\beta_0}(\bar{\pi};V)]$ are both strictly decreasing and continuous in V. By the above, $\tilde{P}'(V) \to 0$ as $V \to \infty$. As $p(\bar{\pi};V) \to 0$ and $m_{\beta_0}(\bar{\pi};V) \to 0$ as $V \to \infty$ for each $\bar{\pi}$, with $p(\bar{\pi};V) \geq m_{\beta_0}(\bar{\pi};V)$, the gap $\bar{\Delta}(V) \to 0$ as well. Therefore, there exists V' such that for all $V \geq V'$,

$$2\bar{\Delta}(V) + \tilde{P}'(V) < \gamma(\eta_2, \varepsilon)^{-1},$$

which is the uniqueness condition of Proposition 7(i). By that proposition, the full model admits a unique steady state $\bar{\pi}_1^*(V) \in [0,1)^2$ for all $V \geq V'$. By Lemma 2, this steady state is not $(1,1)$, so $\bar{\pi}_1^*(V) \in (0,1)$. Moreover, $\bar{\pi}_1^*(V) \leq \bar{\pi}_1^*(V^0) < \bar{\pi}_2^*(V^0)$, proving statement (i).

Now fix $V'' \geq V'$ satisfying statement (i) and let $\bar{\pi}^0 \in \big(\bar{\pi}_2^*(V^0), 1\big)$. Under V'', the unique steady state $\bar{\pi}_1^*(V'') < \bar{\pi}_2^*(V^0)$ is globally attracting by the contraction established in the proof of Proposition 7(i). Hence, the trajectory $\{\bar{\pi}^t\}_{t\geq 0}$ converges to $\bar{\pi}_1^*(V'')$ from $\bar{\pi}^0$. As $\bar{\pi}_1^*(V'') < \bar{\pi}_2^*(V^0)$ and convergence is guaranteed in finite time since the unique steady lies strictly below $\bar{\pi}_2^*(V_0)$, there exists a $t' < \infty$ such that $\bar{\pi}^{t'} < \bar{\pi}_2^*(V^0)$. Returning V to V^0 at time t' restores the original three-steady-state configuration. Because $\bar{\pi}^{t'} < \bar{\pi}_2^*(V^0)$, the trajectory now lies in the basin of attraction of $\bar{\pi}_1^*(V^0)$ under V^0, and converges to $\bar{\pi}_1^*(V^0) < \bar{\pi}^0$, proving statement (ii). □

Proof of Lemma 3.

Proof. Recall that $\Delta\pi_\tau = (1-\eta_\tau)[1 - \bar{\rho}_\tau(\pi) - \pi_\tau] + \eta_\tau(1-\pi_\tau)\varepsilon$. As $\bar{\rho}_\tau(\pi) \in [0,1]$ and $\varepsilon \geq 0$, the expression is minimized at $\bar{\rho}_\tau = 1$ and $\varepsilon = 0$, giving $\Delta\pi_\tau \geq -(1-\eta_\tau)\pi_\tau$. Therefore, $-\Delta\pi_\tau \leq (1-\eta_\tau)\pi_\tau \leq 1 - \eta_\tau$, and

$$\bar{\pi}^t - \bar{\pi}^{t+1} = -(q\Delta\pi_1 + (1-q)\Delta\pi_2) \leq q(1-\eta_1)\pi_1 + (1-q)(1-\eta_2)\pi_2 \leq q(1-\eta_1) + (1-q)(1-\eta_2).$$

Substituting $1-\eta_1 = (1-q)(1+\mu)$ and $1-\eta_2 = q(1+\mu)$ yields $\bar{\pi}^t - \bar{\pi}^{t+1} \leq q(1-q)(1+\mu) + (1-q)q(1+\mu) = 2q(1-q)(1+\mu) = C^*(\mu, q)$. □

Proof of Corollary 3.

Proof. From the proof of Proposition 8, $g(\bar{\pi};\mu,V) \equiv \Psi(\bar{\pi};\mu,V) - \bar{\pi}$ satisfies $\frac{\partial g(\bar{\pi}_2^*)}{\partial \bar{\pi}} > 0$ at the unstable tipping point (where Ψ crosses the diagonal from below). From the proof of Proposition 8, $\frac{\partial g}{\partial \mu} < 0$. From the proof of Proposition 10, $\frac{\partial g}{\partial V} < 0$. Applying the implicit function theorem to $g(\bar{\pi}_2^*;\mu,V) = 0$ in each argument separately yields

$$\frac{d\bar{\pi}_2^*}{d\mu} = -\frac{\partial g/\partial\mu}{\partial g/\partial\bar{\pi}} > 0, \qquad \frac{d\bar{\pi}_2^*}{dV} = -\frac{\partial g/\partial V}{\partial g/\partial\bar{\pi}} > 0. \qquad □$$

Proof of Proposition 11.

Proof. By Proposition 10(i), under (V'', μ^0) the model admits a unique globally attracting steady state $\bar{\pi}_1^*(\mu^0, V'') < \bar{\pi}_2^*(\mu^0, V^0)$. Since the trajectory $\{\bar{\pi}^t\}$ converges to $\bar{\pi}_1^*(\mu^0, V'')$, it eventually falls strictly below both $\bar{\pi}_2^*(\mu^0, V^0)$ and $\bar{\pi}_2^*(\tilde{\mu}, V^0)$, so $T_V, \tilde{T}_V < \infty$.

By hypothesis and Corollary 3, $\bar{\pi}^0 \geq \bar{\pi}_2^*(\tilde{\mu}, V^0) > \bar{\pi}_2^*(\mu^0, V^0)$, so $\tilde{T}_V \geq 1$. By definition of $\tilde{T}_V$ as a minimum, $\bar{\pi}^{\tilde{T}_V} < \bar{\pi}_2^*(\tilde{\mu}, V^0)$ and $\bar{\pi}^{\tilde{T}_V - 1} \geq \bar{\pi}_2^*(\tilde{\mu}, V^0)$.

Suppose for contradiction that $\bar{\pi}^{\tilde{T}_V} \leq \bar{\pi}_2^*(\mu^0, V^0)$. Then

$$\bar{\pi}^{\tilde{T}_V - 1} - \bar{\pi}^{\tilde{T}_V} \geq \bar{\pi}_2^*(\tilde{\mu}, V^0) - \bar{\pi}_2^*(\mu^0, V^0) > C^*(\mu^0, q),$$

where the second inequality is the assumption on $C^*(\mu^0, q)$. Lemma 3 applied under matching parameter μ^0 gives $\bar{\pi}^{\tilde{T}_V - 1} - \bar{\pi}^{\tilde{T}_V} \leq C^*(\mu^0, q)$, a contradiction. Hence, $\bar{\pi}^{\tilde{T}_V} \in \big(\bar{\pi}_2^*(\mu^0, V^0), \bar{\pi}_2^*(\tilde{\mu}, V^0)\big)$. In particular $\bar{\pi}^{\tilde{T}_V} > \bar{\pi}_2^*(\mu^0, V^0)$, so the trajectory has not yet crossed the lower threshold at time $\tilde{T}_V$, establishing $\tilde{T}_V < T_V$.

Under $(\tilde{\mu}, V^0)$, the maintained conditions ensure the three-steady-state structure holds. From the proof of Proposition 8, $\bar{\pi}_1^*(\mu, V)$ is strictly decreasing in μ, so $\bar{\pi}_1^*(\tilde{\mu}, V^0) < \bar{\pi}_1^*(\mu^0, V^0) < \bar{\pi}_2^*(\mu^0, V^0)$. As $\bar{\pi}^{\tilde{T}_V} < \bar{\pi}_2^*(\tilde{\mu}, V^0)$, the trajectory under $(\tilde{\mu}, V^0)$ initiated at $\bar{\pi}^{\tilde{T}_V}$ lies in the basin of attraction of $\bar{\pi}_1^*(\tilde{\mu}, V^0)$ and converges to it. As $\bar{\pi}_1^*(\tilde{\mu}, V^0) < \bar{\pi}_2^*(\mu^0, V^0)$, the trajectory eventually falls strictly below $\bar{\pi}_2^*(\mu^0, V^0)$, so $T_\mu < \infty$.

Under (V'', μ^0), $T_V < \infty$ and by definition $\bar{\pi}^{T_V} < \bar{\pi}_2^*(\mu^0, V^0)$. Restoring V^0 at $t = T_V$ returns the model to the three-steady-state configuration under (μ^0, V^0), with $\bar{\pi}^{T_V}$ lying strictly in the basin of attraction of $\bar{\pi}_1^*(\mu^0, V^0)$. The economy therefore converges persistently to $\bar{\pi}_1^*(\mu^0, V^0) < \bar{\pi}^0$.

As established above, $\bar{\pi}^{\tilde{T}_V} \in \big(\bar{\pi}_2^*(\mu^0, V^0), \bar{\pi}_2^*(\tilde{\mu}, V^0)\big)$. At $t = \tilde{T}_V$, V is restored to V^0 and μ is raised to $\tilde{\mu}$. The maintained conditions ensure three steady states under $(\tilde{\mu}, V^0)$. As $\bar{\pi}^{\tilde{T}_V} < \bar{\pi}_2^*(\tilde{\mu}, V^0)$, the trajectory under $(\tilde{\mu}, V^0)$ lies in the basin of attraction of $\bar{\pi}_1^*(\tilde{\mu}, V^0)$ and converges to it. As $\bar{\pi}_1^*(\tilde{\mu}, V^0) < \bar{\pi}_2^*(\mu^0, V^0)$, the trajectory crosses $\bar{\pi}_2^*(\mu^0, V^0)$ in finite time T_μ, by the argument above. At $t = \tilde{T}_V + T_\mu$, both instruments are restored to (μ^0, V^0). By the definition of T_μ, $\bar{\pi}^{\tilde{T}_V + T_\mu} < \bar{\pi}_2^*(\mu^0, V^0)$, so the trajectory lies strictly in the basin of attraction of $\bar{\pi}_1^*(\mu^0, V^0)$ and converges to it persistently. The strict inequality $\tilde{T}_V < T_V$ is established above, proving the proposition. □

Proof of Corollary 4.

Proof. Sequencing reduces total weighted cost iff $c_V T_V > c_V \tilde{T}_V + c_\mu T_\mu$, i.e., $c_V(T_V - \tilde{T}_V) > c_\mu T_\mu$. □